Web Technologies LAB

(CIE-356P)

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Roll No : 12214802722 **(AIML II-B)**

Semester : 6th Semester

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उद्यमेन हि सिध्यन्ति
कार्याणि न मनोरथैः

MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

Computer Science & Engineering Department

VISION

"To be centre of excellence in education, research and technology transfer in the field of computer engineering and promote entrepreneurship and ethical values."

MISSION

To foster an open, multidisciplinary and highly collaborative research environment for producing world-class engineers capable of providing innovative solutions to real-life problems and fulfil societal needs.

Department of Computer Science & Engineering

VISION

"To attain global excellence through education, innovation, research, and work ethics in the field of Computer Science and engineering with the commitment to serve humanity."

MISSION

M1 To lead in the advancement of computer science and engineering through internationally recognized research and education.

M2 To prepare students for full and ethical participation in a diverse society and encourage lifelong learning.

M3 To foster development of problem solving and communication skills as an integral component of the profession.

M4 To impart knowledge, skills and cultivate an environment supporting incubation, product development, technology transfer, capacity building and entrepreneurship in the field of computer science and engineering.

M5 To encourage faculty, student's networking with alumni, industry, institutions, and other stakeholders for collective engagement

MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

MAIT

PRACTICAL RECORD

PAPER CODE: CIE-356P

Name of the Student: Yash Pandey

Roll No: 12214802722

Branch: Computer Science and Engineering

Group: 6 CSE (AIML-II-B)

S. No.	Experiments	Marks					Total Marks	Remarks	Sign
		R1	R2	R3	R4	R5			
1	Design web pages for your college containing a description of the courses, departments, faculties, library etc, use href, list tags.	2	2	2	2	2	~~08~~ 10		30/1
2	Write html code to develop a webpage having two frames that divide the webpage into two equal rows and then divide the row into equal columns fill each frame with a different background color.	2	2	2	2	2	10 ~~08~~		30/1
3	Design a web page of your home town with an attractive background color, text color, an Image, font etc. (use internal CSS).	2	2	2	2	2	10		13/2
4	Use External, Internal, and Inline CSS to format college web page that you created.								
5	Create HTML. Page with JavaScript which takes Integer number as input and tells whether the number is ODD or EVEN								
6	Create HTML. Page that contains form with fields Name, Email, Mobile No, Gender, Favorite Color and a button now write a JavaScript code to combine and display the information								

	in textbox when the button is clicked and implement validation,								
7	Create XML file to store student information like Enrolment Number, Name Mobile Number, Email Id.								
8	Write a php script to read data from txt file and display it in html table (the file contains info in format Name: Password: Email)								
9	Write a PHP Script for login authentication. Design an html form which takes username and password from user and validate against stored username and password in file.								
10	Write PHP Script for storing and retrieving user information from MySql table.								
11	Write an HTML, CSS and JS Program To Implement Basic Calculator Website								
12	Write an HTML, CSS and JS Program To Imp. Quiz app on Website								

Lab Assessment Sheet

S.No	Experiment	Date	Marks					Total Marks	Signature
			R1	R2	R3	R4	R5		

Lab Assessment Sheet

S.No	Experiment	Date	Marks					Total Marks	Signature
			R1	R2	R3	R4	R5		

Lab Exercise-1. *(23/01/2025)*

Q :- Design Page for your college containing a description of the courses, dept. , faculties, library etc using href, list tags?

A webpage is a digital document displayed on the internet through a web browser, consisting of elements like text, images, videos, and links. It is created using HTML for structure and CSS for styling, making it either static with fixed content or dynamic with interactive features.

1. **HTML (HyperText Markup Language**) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and functional webpages for users.
2. **CSS (Cascading Style Sheets)** is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly

1. Index.html File

```
<!DOCTYPE html>
<html lang="en">
 <head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <link rel="stylesheet" href="style.css" />
  <title>College Website</title>
 </head>
 <body>
  <header style="display: grid; grid-auto-flow: column; align-items: center">
```

```html
  <img
   src="MAIT.png"
   alt="Example Image"
   style="width: 7rem; height: 7rem"
  />
  <h1>Welcome to Maharaja Agrasen Institute of Technology</h1>
</header>
<nav>
  <a href="#home">Home</a>
  <a href="#courses">Courses</a>
  <a href="#departments">Departments</a>
  <a href="#faculty">Faculty</a>
  <a href="#library">Library</a>
</nav>
<div class="container">
  <section id="courses">
   <h2>Courses Offered</h2>
   <p>
    We offer a wide range of undergraduate and postgraduate courses. Some
    of the key courses are:
   </p>
   <ul>
    <li><a href="#Btech">Bachelor of Computer Science</a></li>
    <li><a href="#Arts">Bachelor of Arts in Literature</a></li>
    <li><a href="#BBA">Master of Business Administration</a></li>
    <li><a href="#BSc">Master of Science in Physics</a></li>
   </ul>
  </section>
```

```html
<section id="departments">
 <h2>Departments</h2>
 <p>Our college is home to the following departments:</p>
 <ul>
  <li><a href="#comp-sci">Computer Science</a></li>
  <li><a href="#arts">Arts</a></li>
  <li><a href="#physics">Physics</a></li>
  <li><a href="#business">Business Administration</a></li>
 </ul>
</section>
<section id="faculty">
 <h2>Faculty</h2>
 <p>Meet our esteemed faculty members who provide quality education:</p>
 <ul>
  <li>
   <a href="#CSE-HOD">Dr. John Doe - Head of Computer Science</a>
  </li>
  <li>
   <a href="#Arts-HOD">Prof. Jane Smith - Department of Literature</a>
  </li>
  <li>
   <a href="#Bsc-HOD">Dr. Emily White - Department of Physics</a>
  </li>
  <li><a href="#BBA-HOD">Prof. Richard Brown - MBA Program</a></li>
 </ul>
</section>
<section id="library">
 <h2>Library</h2>
```

```html
      <p>
        Our library offers a vast collection of books, journals, and research
        papers for all students and staff. For more information:
      </p>
      <ul>
        <li><a href="#library1">Visit the Library Website</a></li>
        <li><a href="#library2">Library Timings</a></li>
        <li><a href="#library3">Borrowing Policy</a></li>
      </ul>
    </section>
  </div>
  <footer>
    <p>&copy; 2025 MAIT. All rights reserved.</p>
    <p>Email: mait@college.edu</p>
    <p>Phone: (+91) 9130836542</p>
  </footer>
 </body>
</html>
```

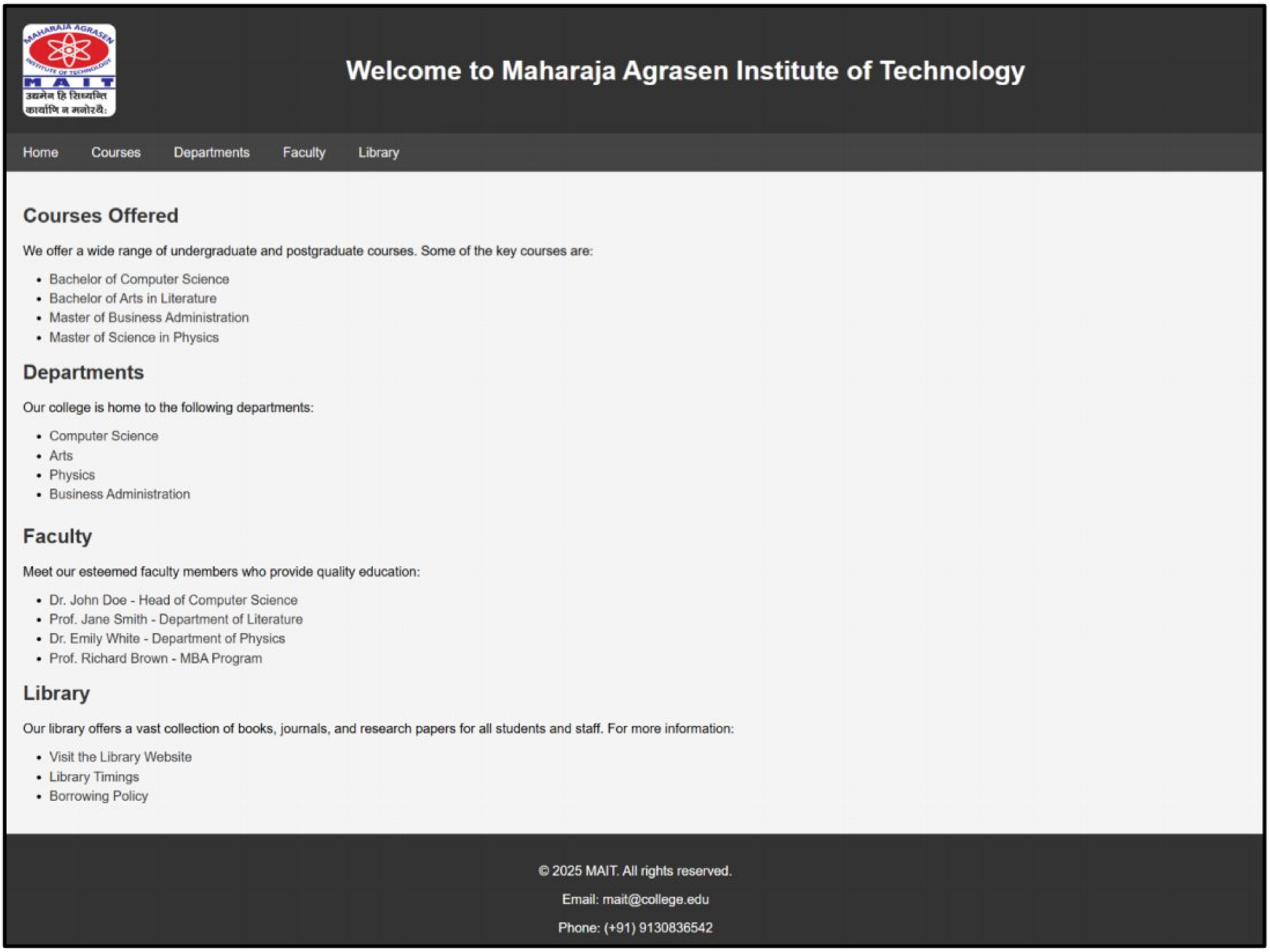
2. Style.css File

```css
body {
  margin: 0;
  padding: 0;
}
img{
  margin: 5px;
  display: inline;
  border-radius: 10px;
}
```

```
nav {
  background-color: #444;
  overflow: hidden;
  color: white;
  padding: 14px 20px;
}
nav a:hover {
  background-color: #ddd;
  color: black;
}
h2 {
  color: #333;
}
ul {
  list-style-type: disc;
  padding-left: 2rem;
}
li {
  margin: 5px 0;
}
footer {
  background-color: #333;
  color: white;
  padding: 20px 0;
  text-align: center;
}
```

```css
.footer-content {
  display: flex;
  justify-content: space-evenly;
}
a{
  color: #333;
  text-decoration: none;
}
```

OUTPUT : -



Viva Questions (Lab Exercise -1)

1. **Can you explain how you would use href attribute in HTML to create hyperlinks for navigating to different sections of your college's website?**

The href attribute in the <a> tag is used to create clickable links. It specifies the URL or section to navigate to. For example, About Us links to the About page of the college website.

2. **How would you structure the HTML code to list the courses offered by your college on the website?**

I would use <ul> (unordered list) or <ol> (ordered list) with <li> tags to display the courses. Each course name is written inside <li>, providing a clear and readable structure for visitors.

3. **Describe how you would use the id attribute in HTML to create a link that navigates to a specific section of the webpage.**

First, assign an id to the target section, like <div id="courses">. Then, use an anchor tag with href="#courses" to link to it. Clicking the link scrolls the page to that specific section.

4. **Explain how you would use the list tags in HTML to display information about the faculties in your college.**

I would use a list structure such as <ul> with <li> tags to show faculty details. Each list item can contain the faculty's name and department for easy viewing and neat presentation.

5. **How would you use the id attribute to create a link that navigates to the "Library" section of your college's website?**

I would give the Library section an ID like <section id="library">. Then, I'd create a link using Library so that clicking it scrolls directly to the Library content.

Lab Exercise-2. *(30/01/2025)*

Q :- Write html code to develop a webpage having 4 Frames With 2 Row And Columns Along with a different background color?

A webpage is a digital document displayed on the internet through a web browser, consisting of elements like text, images, videos, and links. It is created using HTML for structure and CSS for styling, making it either static with fixed content or dynamic with interactive features.

1. **HTML (HyperText Markup Language**) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and functional webpages for users.
2. **CSS (Cascading Style Sheets)** is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly

1. Index.html File

```
<!DOCTYPE html>

<html lang="en">

 <head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <link rel="stylesheet" href="style.css" />

  <title>Webpage with Frames</title>

 </head>

 <body>

  <h1>Frames with Different Background Color</h1>

  <div class="frames">
```

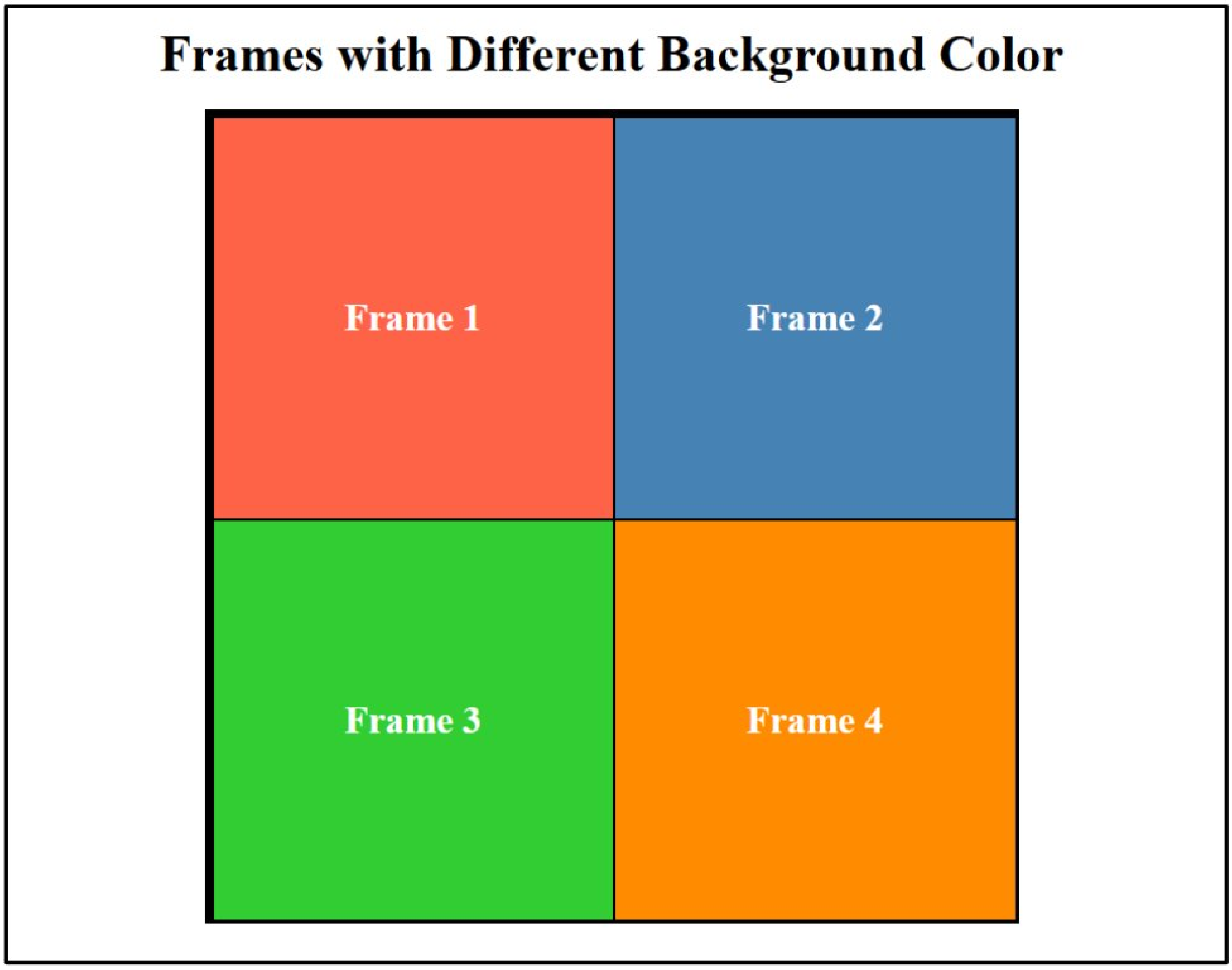
```html
      <div class="row">
        <div class="frame frame1">Frame 1</div>
        <div class="frame frame2">Frame 2</div>
      </div>
      <div class="row">
        <div class="frame frame3">Frame 3</div>
        <div class="frame frame4">Frame 4</div>
      </div>
    </div>
  </body>
</html>
```

2. style.css File

```css
body {
  margin: 0;        display: flex;
  flex-direction: column;           align-items: center;
  height: 100vh; /* Full viewport height */
}
h1 {
  margin: 5rem;             margin-bottom: 1rem;
}
.frames {
  height: 500px;            width: 500px;
  font-size: 1.5rem;        font-weight: bold;
  color: white;    border: 5px solid black;
}
.row {
  display: grid;
  grid-template-columns: 1fr 1fr;
}
```

```
.frame1 {
  background-color: #ff6347; /* Tomato */
}
.frame2 {
  background-color: #4682b4; /* Steel Blue */
}
.frame3 {
  background-color: #32cd32; /* Lime Green */
}
.frame4 {
  background-color: #ff8c00; /* Dark Orange */
}
```

OUTPUT : -



Viva Questions (Lab Exercise -2)

1. **How would you use HTML to create a webpage with two frames that divide the page into two equal rows?**

We can use two <div> elements inside a parent container and apply CSS to give each div 50% height. For layout, we can use Flexbox or simple CSS styling with height: 50% and width: 100% to divide the page into two equal horizontal sections.

2. **How can you divide each row into equal columns using HTML and CSS?**

Inside each row <div>, we can add multiple column <div> elements and use CSS Flexbox with display: flex; to distribute them evenly. Setting flex: 1; on each column ensures equal width distribution within the row.

3. **Describe how you would fill each frame with a different background color using CSS.**

Assign a unique class or ID to each frame <div>, then use CSS to apply different background colors. For example, .frame1 { background-color: lightblue; } and .frame2 { background-color: lightgreen; } will give each frame a distinct color.

4. **How can you ensure that each row is divided equally on the webpage?**

To divide the page into equal rows, we set both rows' height to 50% using CSS. Also, we ensure the parent container has height: 100vh (full viewport height), so both rows take equal space vertically.

5. **Explain how the display: flex; property works in CSS to create equal columns within a row.**

The display: flex; property allows elements to be arranged in a flexible row or column layout. When used on a row container, setting flex: 1; on each child divides the space equally among columns, maintaining responsiveness.

Lab Exercise-3. *(06/02/2025)*

Q :- Design a web page of your home town with attractive color, Text, Image & font?

A webpage is a digital document displayed on the internet through a web browser, consisting of elements like text, images, videos, and links. It is created using HTML for structure and CSS for styling, making it either static with fixed content or dynamic with interactive features.

1. **HTML (HyperText Markup Language**) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and functional webpages for users.
2. **CSS (Cascading Style Sheets)** is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly

1. Index.html File

```
<!DOCTYPE html>
<html lang="en">
 <head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <link rel="stylesheet" href="style.css" />
  <title>Welcome to Rewa - The Land of White Tigers</title>
 </head>
 <body>
  <header>
   <h1>Welcome to Rewa, Madhya Pradesh</h1>
   <p>The Land of White Tigers and Rich Heritage</p>
```

```
  </header>
  <section class="hero">
    <p>
      Rewa is famously known as the <b>Land of White Tigers</b> because the first white tiger in the world was discovered here. The rare white tiger, named Mohan, was found in the jungles of Rewa in 1951 by <b>Maharaja Martand Singh</b>. This discovery played a crucial role in the global conservation and breeding efforts for white tigers. Today, Rewa stands as a proud symbol of this magnificent species and continues to be associated with wildlife heritage, attracting nature lovers and researchers from around the world.
    </p>
    <br />
    <div style="display: flex; justify-content: center; gap: 15px">
      <img src="Images/Mukundpur.png" alt="White Tiger of Rewa" style="width: 30%; height: auto; border-radius: 10px" />
      <img src="Images/White_Tiger_of_Rewa.png" alt="White Tiger of Rewa" style="width: 30%; height: auto; border-radius: 10px" />
    </div>
  </section>
  <section class="content">
    <div class="card">
      <img src="Images/Rewa_Fort.png" alt="Rewa Fort" height="200px" width="150px" />
      <h2>Rewa Fort</h2>
      <p>An architectural marvel with a deep history, Rewa Fort is a must-visit for history enthusiasts.</p>
    </div>
    <div class="card">
      <img src="Images/Keoti_Waterfall.png" alt="Keoti Waterfall" height="200px" width="150px" />
      <h2>Keoti Waterfall</h2>
      <p>Keoti Waterfall, one of the highest waterfalls in Madhya Pradesh, is a breathtaking sight during monsoons.</p>
    </div>
    <div class="card">
```

```
        <img src="Images/Rani_Talab.png" alt="Rani Talab" height="200px" width="150px" />

        <h2>Rani Talab</h2>

        <p>One of the oldest and holiest water bodies in Rewa, Rani Talab is a serene place for visitors.</p>

      </div>

    </section>

    <footer>

      <p>&copy; 2025 Yash Pandey | Explore the Beauty of Rewa</p>

      <br />

      <p>

        <a href="https://github.com/YashPandey1405/" target="_blank"><i class="fab fa-github"></i></a>

        <a href="https://www.linkedin.com/in/yashpandey29/" target="_blank"><i class="fab fa-linkedin"></i></a>

      </p>

    </footer>

  </body>

</html>
```

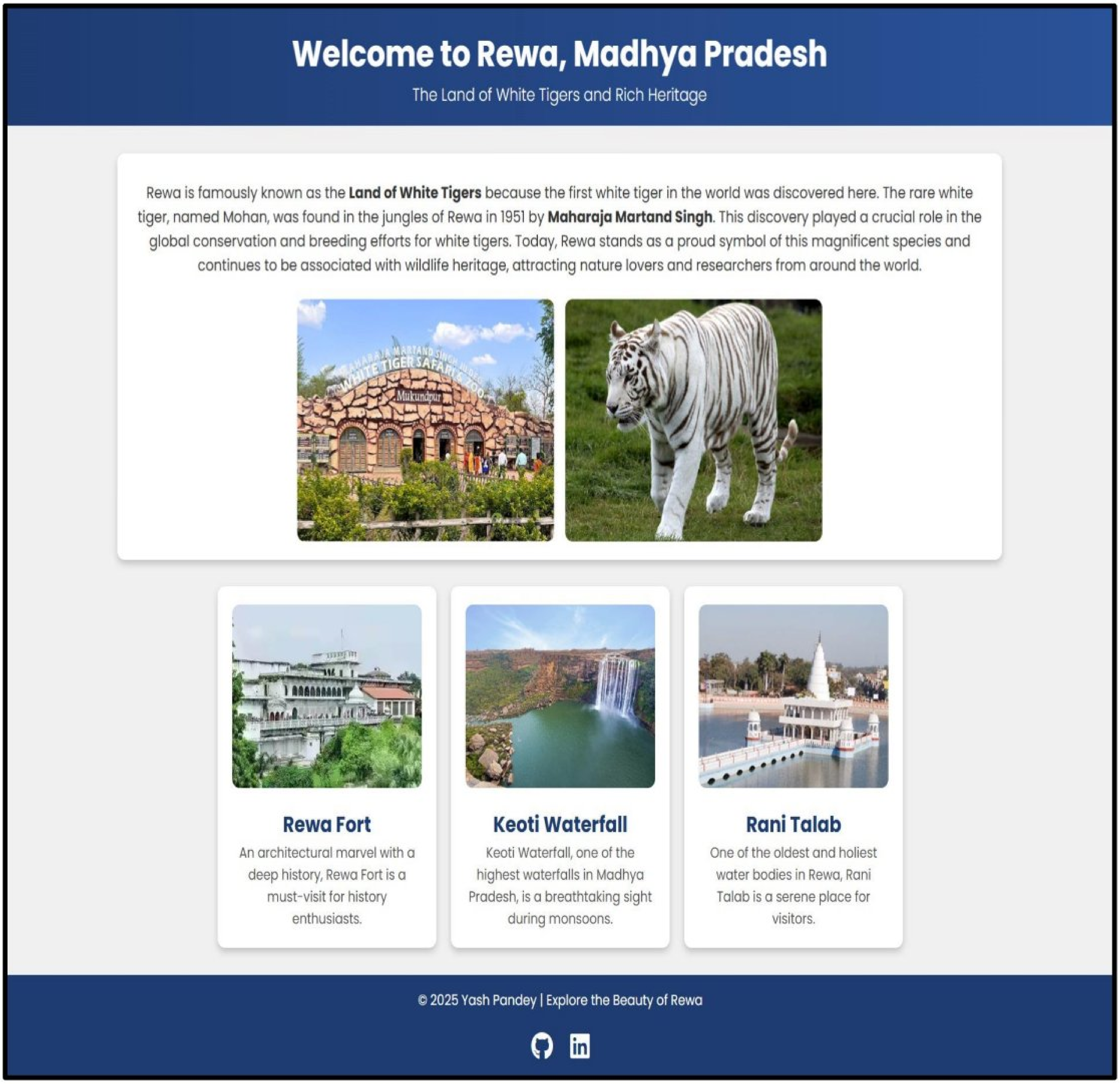
2. style.css File

```
body {

  background-color: #f0f0f0;

  color: #333;

}

header {

  background: linear-gradient(to right, #1e3c72, #2a5298);

  color: white;

  text-align: center;

  padding: 20px;

}
```

```css
header h1 {
  font-size: 2.5em;
  font-size: 1.2em;
  font-weight: 300;
}
.hero {
  text-align: center;
  margin: 30px 10%;
  padding: 20px;
  background-color: white;
  border-radius: 10px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
  margin-top: 10px;
  font-size: 1.1em;
}
.content {
  display: flex;
  flex-wrap: wrap;
  justify-content: center;
  gap: 20px;
  margin: 30px 10%;
}
.card {
  background: white;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
  width: 300px;            text-align: center;
}
```

```
footer {
  background: #1e3c72;
  color: white;
  text-align: center;
  padding: 15px;
  margin-top: 20px;
}
```

OUTPUT : -



Viva Questions (Lab Exercise -3)

1. **How would you use internal CSS to set an attractive background color for your hometown's web page?**

I would use the <style> tag inside the <head> section of the HTML file and target the body tag. For example, body { background-color: lightblue; } will apply a soft blue background to the entire page.

2. **Describe how you would use internal CSS to set a different text color for headings and paragraphs on the web page.**

Within the <style> tag, I would use selectors like h1 or h2 for headings and p for paragraphs. For example, h1 { color: navy; } p { color: darkgray; } sets distinct text colors for each.

3. **How can you use internal CSS to add an image to your hometown's web page?**

Using internal CSS, I can add a background image by targeting a section or the body. For example, body { background-image: url('hometown.jpg'); background-size: cover; } adds the image and covers the full screen.

4. **Explain how you would use internal CSS to set a custom font for the text on your hometown's web page.**

Inside the <style> tag, I can apply a font using font-family. For example, body { font-family: 'Verdana', sans-serif; } changes the default text font across the webpage to a more custom, clean look.

5. **How would you use internal CSS to add styling to a navigation menu on your hometown's web page?**

I'd define styles for a <nav> or <ul> element in the <style> tag, such as nav { background-color: #333; } nav a { color: white; padding: 10px; text-decoration: none; } to style the menu bar attractively.

Lab Exercise-4. *(13/02/2025)*

Q :- Design Page for your college containing a description of the courses, dept. , faculties, library etc using href, list tags Using External, Internal, and Inline CSS ?

HTML (HyperText Markup Language) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and webpages for users.

CSS (Cascading Style Sheets) is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly . It can be applied to a webpage using three methods: **External, Internal, and Inline CSS**.

1. **External CSS** is written in a separate file with a .css extension and linked to the HTML document. It is useful for maintaining a consistent design across multiple pages and allows for easy modifications.
2. **Internal CSS** is placed within the <style> tag inside the <head> section of an HTML file. It is used when styles need to be applied to a single webpage without affecting others.
3. **Inline CSS** is applied directly within an HTML element using the style attribute. It is useful for making quick changes but is not ideal for larger projects due to difficulty in maintenance.

1. Index.html File

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
```

```html
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <link rel="stylesheet" href="style.css" />
  <title>College Website</title>
</head>
<body>
  <header style="display: grid; grid-auto-flow: column; align-items: center">

    <img
      src="MAIT.png"
      alt="Example Image"
      style="width: 7rem; height: 7rem"
    />
    <h1>Welcome to Maharaja Agrasen Institute of Technology</h1>
  </header>
  <nav>
    <a href="#home">Home</a>
    <a href="#courses">Courses</a>
    <a href="#departments">Departments</a>
    <a href="#faculty">Faculty</a>
    <a href="#library">Library</a>
  </nav>
  <div class="container">
    <section id="courses">
      <h2>Courses Offered</h2>
      <p>
        We offer a wide range of undergraduate and postgraduate courses. Some
        of the key courses are:
      </p>
      <ul>
```

```html
    <li><a href="#Btech">Bachelor of Computer Science</a></li>
    <li><a href="#Arts">Bachelor of Arts in Literature</a></li>
    <li><a href="#BBA">Master of Business Administration</a></li>
    <li><a href="#BSc">Master of Science in Physics</a></li>
  </ul>
</section>

<section id="departments">
  <h2>Departments</h2>
  <p>Our college is home to the following departments:</p>
  <ul>
    <li><a href="#comp-sci">Computer Science</a></li>
    <li><a href="#arts">Arts</a></li>
    <li><a href="#physics">Physics</a></li>
    <li><a href="#business">Business Administration</a></li>
  </ul>
</section>
<section id="faculty">
  <h2>Faculty</h2>
  <p>Meet our esteemed faculty members who provide quality education:</p>
  <ul>
    <li>
      <a href="#CSE-HOD">Dr. John Doe - Head of Computer Science</a>
    </li>
    <li>
      <a href="#Arts-HOD">Prof. Jane Smith - Department of Literature</a>
    </li>
    <li>
      <a href="#Bsc-HOD">Dr. Emily White - Department of Physics</a>
```

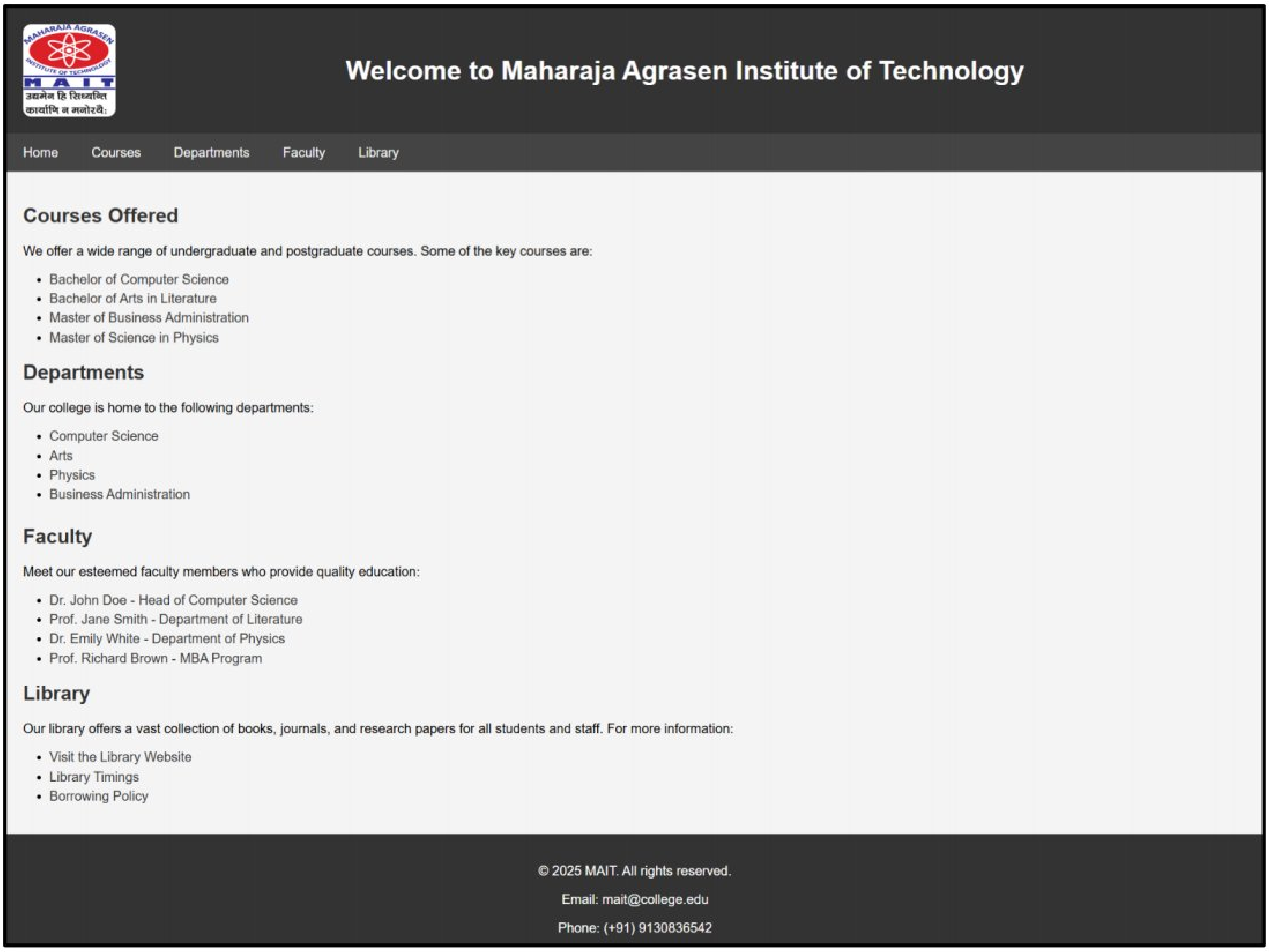
```html
        </li>
        <li><a href="#BBA-HOD">Prof. Richard Brown - MBA Program</a></li>
      </ul>
    </section>
    <section id="library">
      <h2>Library</h2>

      <p>
        Our library offers a vast collection of books, journals, and research
        papers for all students and staff. For more information:
      </p>
      <ul>
        <li><a href="#library1">Visit the Library Website</a></li>
        <li><a href="#library2">Library Timings</a></li>
        <li><a href="#library3">Borrowing Policy</a></li>
      </ul>
    </section>
  </div>
  <footer>
    <p>&copy; 2025 MAIT. All rights reserved.</p>
    <p>Email: mait@college.edu</p>
    <p>Phone: (+91) 9130836542</p>
  </footer>
 </body>
</html>
```

2. Style.css File

```css
body {
  margin: 0;
  padding: 0;
}
img{
  margin: 5px;
  display: inline;
  border-radius: 10px;
}
nav {
  background-color: #444;
  overflow: hidden;
  color: white;
  padding: 14px 20px;
}
nav a:hover {
  background-color: #ddd;
  color: black;
}
ul {
  list-style-type: disc;
  padding-left: 2rem;
}
li {
  margin: 5px 0;
}
footer {
  background-color: #333;
```

```
    color: white;
    padding: 20px 0;
    text-align: center;
}
.footer-content {
    display: flex;
    justify-content: space-evenly;
}
a{
    color: #333;
    text-decoration: none;
}
```

OUTPUT : -



Viva Questions (Lab Exercise -4)

1. **Can you explain how you would use href attribute in HTML to create hyperlinks for navigating to different sections of your college's website?**

The href attribute in the <a> tag is used to create clickable links. It specifies the URL or section to navigate to. For example, About Us links to the About page of the college website.

2. **How would you structure the HTML code to list the courses offered by your college on the website?**

I would use <ul> (unordered list) or <ol> (ordered list) with <li> tags to display the courses. Each course name is written inside <li>, providing a clear and readable structure for visitors.

3. **Describe how you would use the id attribute in HTML to create a link that navigates to a specific section of the webpage.**

First, assign an id to the target section, like <div id="courses">. Then, use an anchor tag with href="#courses" to link to it. Clicking the link scrolls the page to that specific section.

4. **Explain how you would use the list tags in HTML to display information about the faculties in your college.**

I would use a list structure such as <ul> with <li> tags to show faculty details. Each list item can contain the faculty's name and department for easy viewing and neat presentation.

5. **How would you use the id attribute to create a link that navigates to the "Library" section of your college's website?**

I would give the Library section an ID like <section id="library">. Then, I’d create a link using Library so that clicking it scrolls directly to the Library content.

Lab Exercise-5. *(13/02/2025)*

Q :- Create HTML Page with JavaScript which takes Integer number as input and tells whether the number is ODD or EVEN?

A webpage is a digital document displayed on the internet through a web browser, consisting of elements like text, images, videos, and links. It is created using HTML for structure and CSS for styling, making it either static with fixed content or dynamic with interactive features.

1. **HTML (HyperText Markup Language**) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and functional webpages for users.
2. **CSS (Cascading Style Sheets)** is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly .
3. **JavaScript (JS)** is a powerful, lightweight programming language used to make web pages interactive. It enables dynamic content, animations, form validations, and real-time updates without needing to reload the page as a key component of web development .

1. Index.html File

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <link rel="stylesheet" href="style.css" />
    <title>Odd or Even Check</title>
  </head>
```

```html
 <body>
  <header>
   <h1>Hi, I'm Yash Pandey</h1>
   <p style="color: wheat">Welcome to the Odd/Even Number Checker!</p>
  </header>
  <main>
   <section class="container">
    <h2>Odd or Even Checker</h2>
    <p>Enter an integer number:</p>
    <input type="text" id="numberInput" placeholder="Enter a number" />
    <button onclick="checkOddOrEven()">Check</button>
    <p id="result"></p>
   </section>
  </main>
  <script src="script.js" defer></script>
 </body>
</html>
```

2. <u>style.css File</u>

```css
body {
 background-color: #f0f0f0;
 font-family: "Arial", sans-serif;
 display: flex;
 justify-content: center;
 align-items: center;
 min-height: 100vh;
 flex-direction: column;
}
```

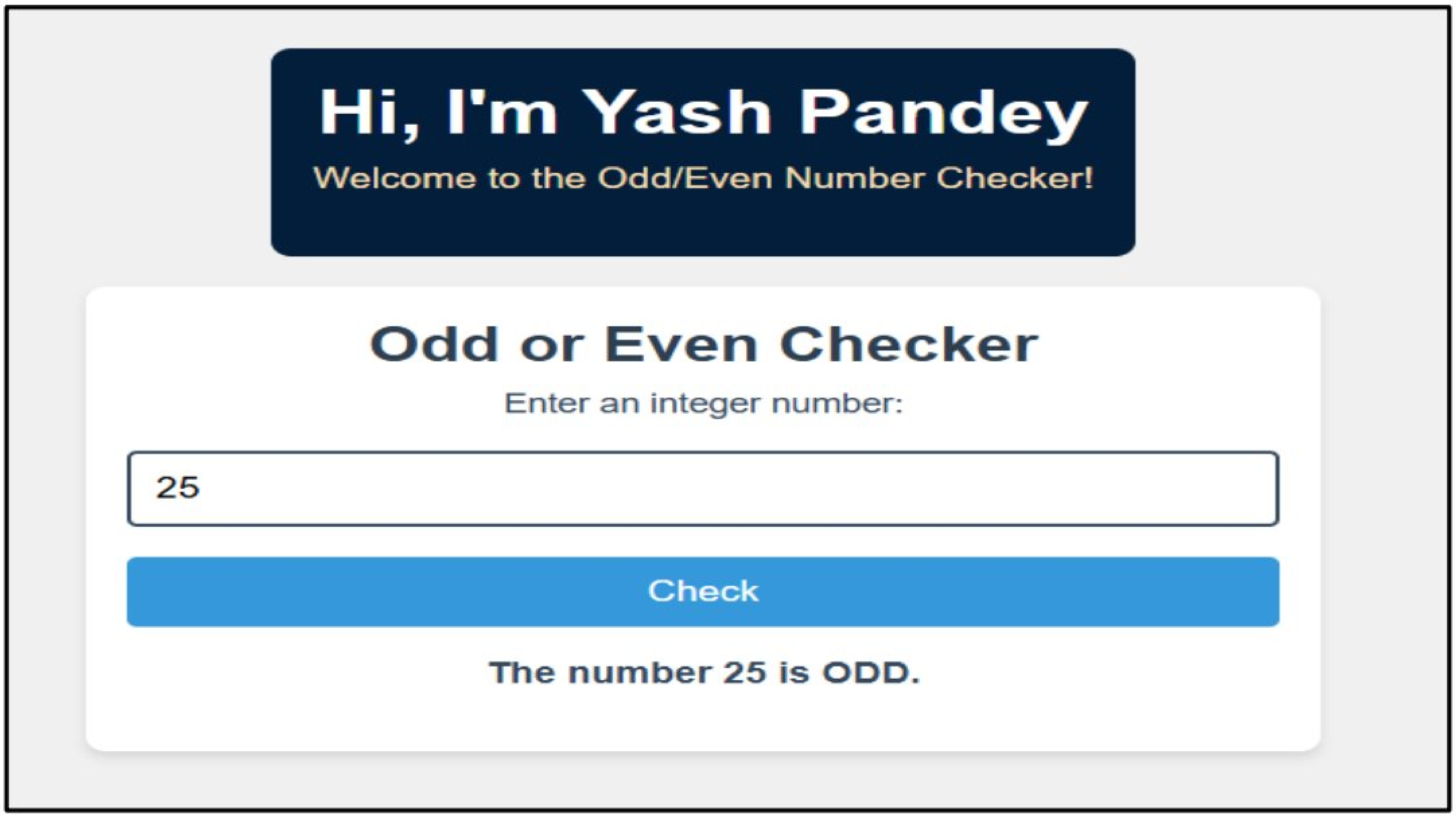
```
header {
 background-color: #031e3a;
 color: white;
 text-align: center;
 padding: 20px;
 border-radius: 10px;
 margin-bottom: 20px;
}
main {
 width: 100%;
 max-width: 600px;
 padding: 20px;
 background-color: white;
 border-radius: 10px;
 box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
}
input {
 width: 100%;
 padding: 12px;
 margin-bottom: 20px;
 border: 2px solid #34495e;
 border-radius: 5px;
 font-size: 1.2em;
}
#result {
 text-align: center;
 font-size: 1.2em;      margin-top: 20px;
 font-weight: bold;
}
```

```css
/* Result success and error styles */
#result.success {
  color: #2ecc71;
}
#result.error {
  color: #e74c3c;
}
```

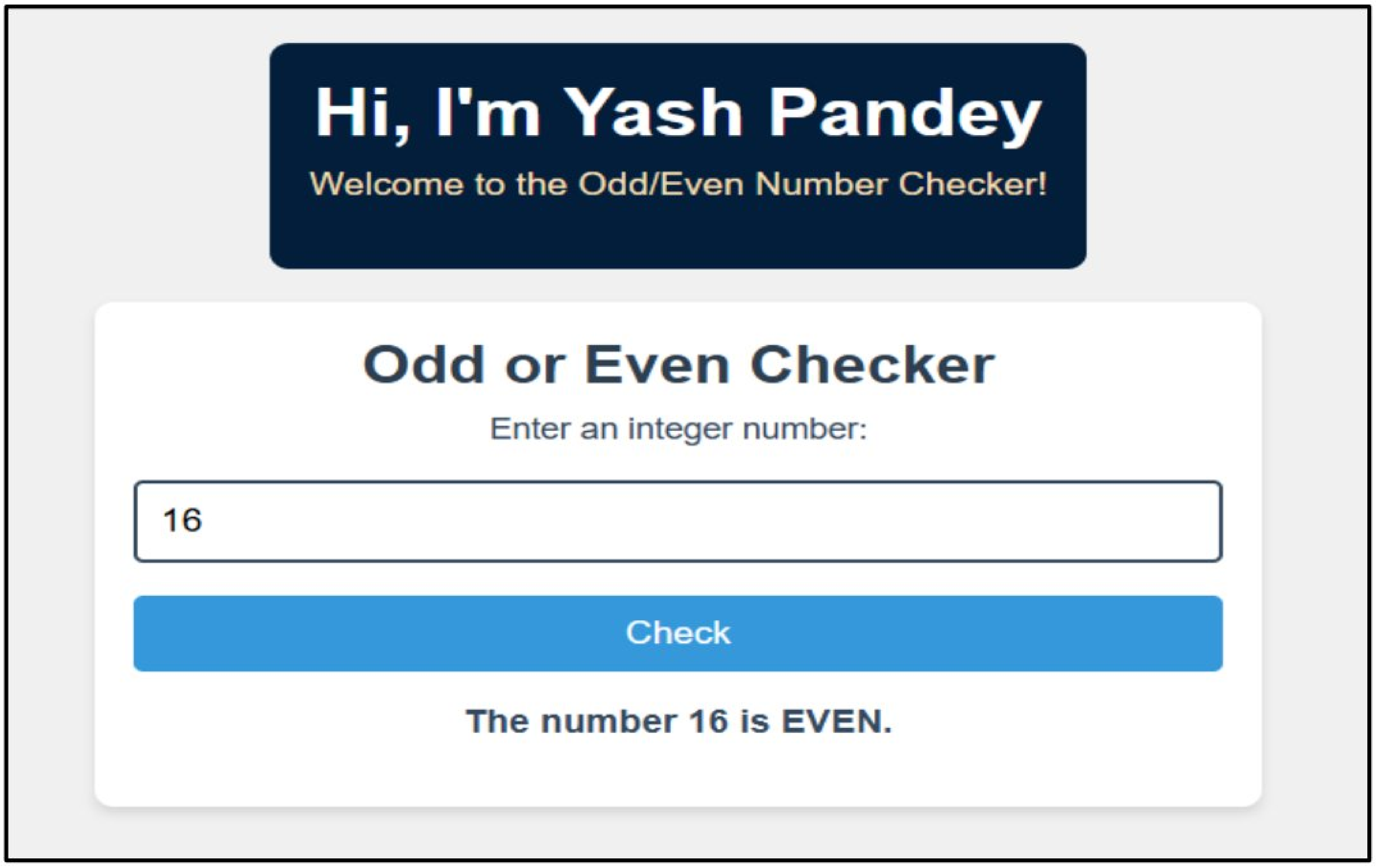
3. script.js File

```javascript
function checkOddOrEven() {
  // Get the number entered by the user
  var number = parseInt(document.getElementById("numberInput").value);
  if (isNaN(number)) {
    document.getElementById("result").innerHTML =
      "Please enter a valid number.";
  } else {
    // Check if the number is even or odd
    if (number % 2 === 0) {
      document.getElementById("result").innerHTML =
        "The number " + number + " is EVEN.";
    } else {
      document.getElementById("result").innerHTML =
        "The number " + number + " is ODD.";
    }
  }
}
```

OUTPUT (For Odd Number) : -



OUTPUT (For Even Number) : -



Viva Questions (Lab Exercise -5)

1. How would you create an HTML form to take an integer input from the user?

I would use the <form> tag with an <input> field of type number. For example: <input type="number" id="userInput"> ensures the user can only input numbers, including integers.

2. Describe how you would use JavaScript to check if the entered number is odd or even.

In JavaScript, I would use the modulo operator %. For example: if (num % 2 === 0) means the number is even, otherwise it's odd. This logic helps check and respond accordingly.

3. How can you ensure that the user enters an integer and not a decimal or a non-numeric value?

You can use Number.isInteger() in JavaScript to validate the input. Also, setting the input type to number and using the step="1" attribute helps prevent decimal input at the HTML level.

4. How would you modify the JavaScript function to display the result on the webpage instead of using an alert box?

Instead of alert(), I'd use document.getElementById("result").innerText = "Your number is even"; to show the message in a specific <div> or <p> element directly on the webpage.

5. How would you handle the case where the user enters a non-integer value or leaves the input field empty?

In JavaScript, I'd first check if the input is empty or isNaN(value), and then use Number.isInteger() to ensure it's a whole number. If not valid, I'd show an error message using inner text.

Lab Exercise-6. *(20/02/2025)*

Q :- Create HTML Page that contains form with fields Name, Email, Mobile No, Gender , Fav. Colour and a button To Show JS code to display the information in textbox ?

A webpage is a digital document displayed on the internet through a web browser, consisting of elements like text, images, videos, and links. It is created using HTML for structure and CSS for styling, making it either static with fixed content or dynamic with interactive features.

1. **HTML (HyperText Markup Language**) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and functional webpages for users.
2. **CSS (Cascading Style Sheets)** is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly
3. **JavaScript (JS)** is a powerful, lightweight programming language used to make web pages interactive. It enables dynamic content, animations, form validations, and real-time updates without needing to reload the page as a key component of web development .

- ❖ When a web page loads, the browser first processes the **HTML** (HyperText Markup Language), which defines the structure and content of the page. Then, the **CSS** (Cascading Style Sheets) is applied to style and format the elements, controlling layout, colors, fonts, and responsiveness. Finally, **JavaScript** (JS) executes by handling events, animations, and updates.
- ❖ The browser parses and renders these components sequentially, ensuring a smooth user experience by integrating structure, design, and functionality into a cohesive webpage. This execution follows a sequence where HTML forms the foundation, CSS enhances visual appeal, and JavaScript brings dynamic behavior.

1. Index.html File

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Personal Information Form</title>
 <link rel="stylesheet" href="styles.css">
</head>
<body class="bg-light">
 <div class="container mt-5">
  <h1 class="text-center mb-4">Personal Information Form</h1>
  <form id="personalForm" novalidate>
   <div class="row mb-3">
    <div class="col-md-6">
     <label for="name" class="form-label">Name:</label>
     <input type="text" id="name" name="name" class="form-control" placeholder="Enter your name" required>
     <div class="invalid-feedback">Please Enter Your Name</div>
    </div>
    <div class="col-md-6">
     <label for="email" class="form-label">Email:</label>
     <input type="email" id="email" name="email" class="form-control" placeholder="Enter your email" required>
     <div class="invalid-feedback">Please Enter a Valid Email</div>
    </div>
   </div>
   <div class="row mb-3">
    <div class="col-md-6">
     <label for="mobile" class="form-label">Mobile No:</label>
```

```
        <input type="tel" id="mobile" name="mobile" class="form-control"
placeholder="Enter your mobile number" required>
        <div class="invalid-feedback">Please Enter a Valid Mobile Number</div>
      </div>
      <div class="col-md-6">
        <label class="form-label">Gender:</label><br>
        <input type="radio" id="male" name="gender" value="Male" class="form-check-input"
required>
        <label for="male" class="form-check-label">Male</label>
        <input type="radio" id="female" name="gender" value="Female" class="form-check-
                                                      input" required>
        <label for="female" class="form-check-label">Female</label>
        <div class="invalid-feedback">Please Select Your Gender</div>
      </div>
    </div>
    <div class="mb-3">
      <label for="color" class="form-label">Favourite Colour:</label>
      <input type="text" id="color" name="color" class="form-control" placeholder="Enter
                                   your favourite colour" required>
      <div class="invalid-feedback">Please Enter Your Favourite Colour</div>
    </div>
    <button type="submit" class="btn btn-outline-danger mt-3">Submit</button>
  </form>
  <div class="mt-4 mb-4">
    <h3>Form Information:</h3>
    <textarea id="result" rows="5" class="form-control" readonly></textarea>
  </div>        </div>
 <script src="script.js"></script>
</body>
</html>
```

2. style.css File

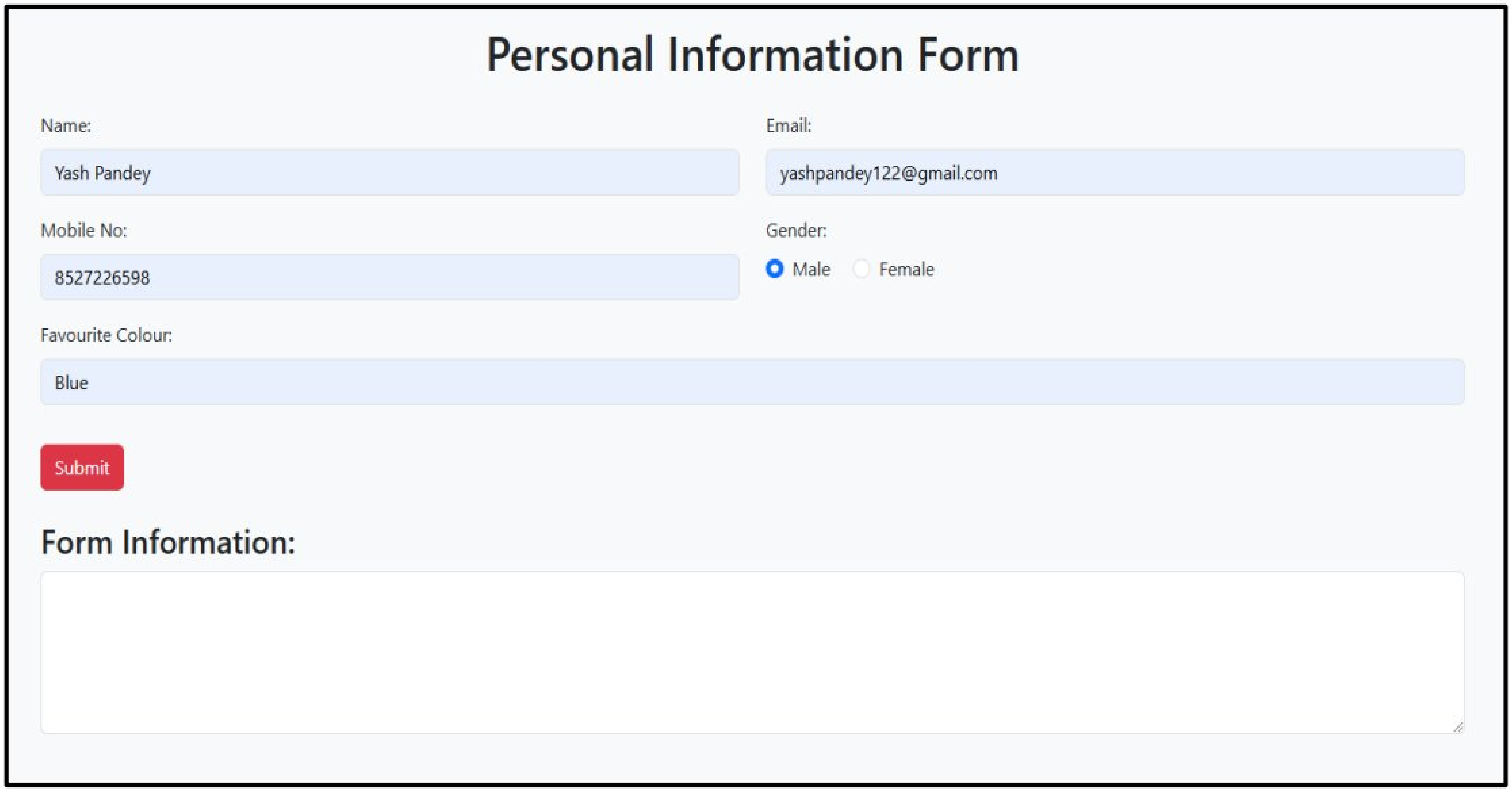
```css
body {
  background-color: #f0f0f0;
  color: #333;
}
header {
  background: linear-gradient(to right, #1e3c72, #2a5298);
  color: white;
  text-align: center;
  padding: 20px;
}
header h1 {
  font-size: 2.5em;
  font-size: 1.2em;
  font-weight: 300;
}
.hero {
  text-align: center;
  margin: 30px 10%;
  padding: 20px;
  background-color: white;
  border-radius: 10px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
  margin-top: 10px;
  font-size: 1.1em;
}
.content {
  display: flex;
  flex-wrap: wrap;
```

```
  justify-content: center;
  gap: 20px;
  margin: 30px 10%;
}
.card {
  background: white;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
  width: 300px;             text-align: center;
}
footer {
  background: #1e3c72;
  color: white;
  text-align: center;
  padding: 15px;
  margin-top: 20px;
}
textarea {
    font-family: Arial, sans-serif;
    resize: none;
}
.invalid-feedback {
    font-size: 0.9rem;
}
h1{
   font-size: 2.5em;        font-size: 1.2em;
  font-weight: 300;
}
```

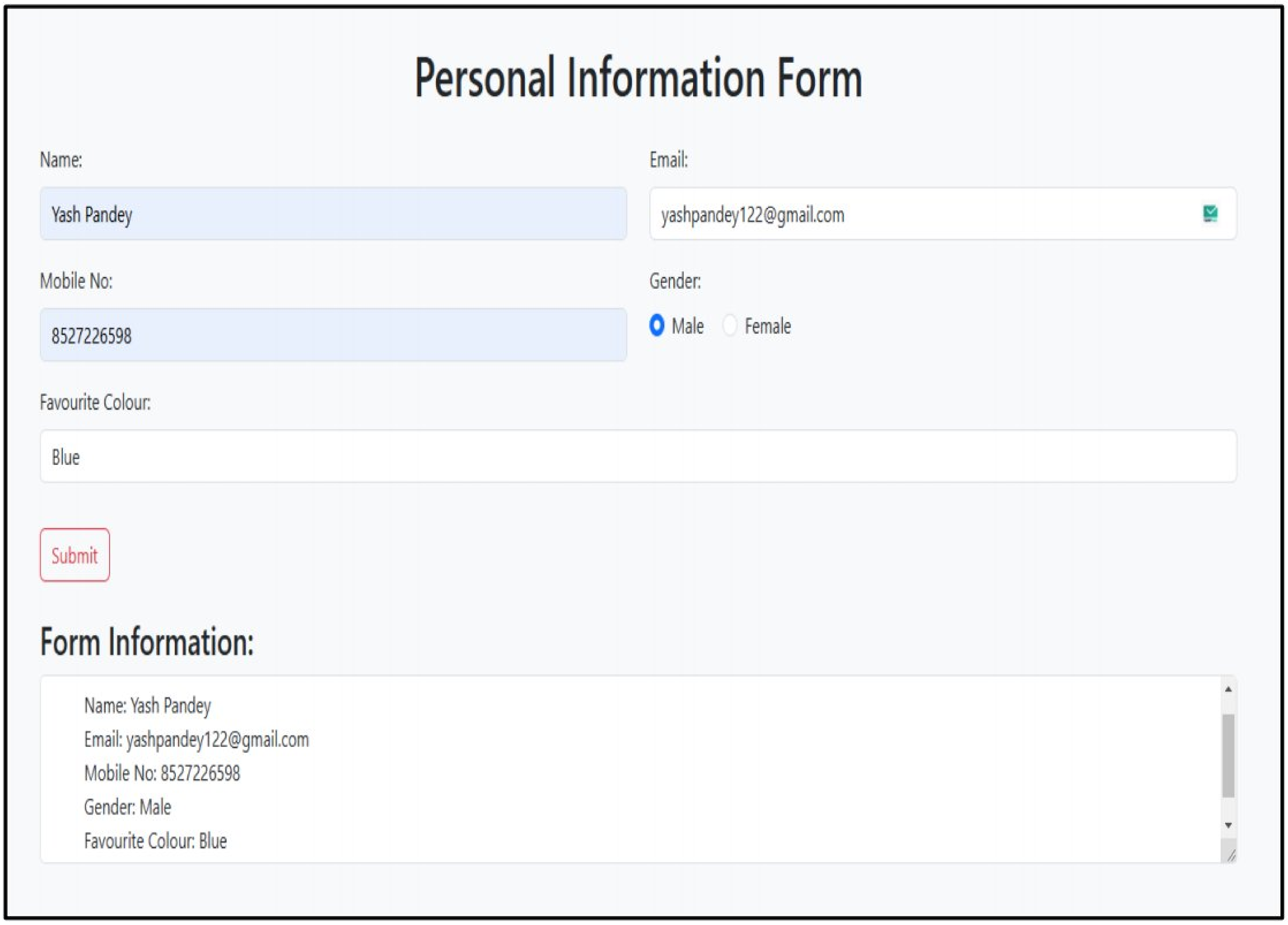
3. script.js File

```
document.getElementById("personalForm").addEventListener("submit", function(event) {
  event.preventDefault(); // Prevent form from submitting

  // Get the values from the form fields
  const name = document.getElementById("name").value;
  const email = document.getElementById("email").value;
  const mobile = document.getElementById("mobile").value;
  const gender = document.querySelector('input[name="gender"]:checked');
  const color = document.getElementById("color").value;

  // Validate form fields
  if (!name || !email || !mobile || !gender || !color) {
    alert("Please fill out all fields!");
    return;
  }
  // Combine the information into a string
  const genderValue = gender ? gender.value : "";
  const resultText = `
    Name: ${name}
    Email: ${email}
    Mobile No: ${mobile}
    Gender: ${genderValue}
    Favourite Colour: ${color}
  // Display the result in the textarea
  document.getElementById("result").value = resultText;
});
```

OUTPUT (Before Clicking Submit Button) :-



OUTPUT (After Clicking Submit Button) : -



Viva Questions (Lab Exercise -6)

1. What HTML elements are used to create the form fields?

HTML form fields are created using elements like <input>, <textarea>, <select>, and <label>. Each <input> tag can have different type attributes like text, email, number, or password to collect specific types of data.

2. What JavaScript function is called when the button is clicked?

The JavaScript function is usually called using the onclick attribute in the button element. For example, <button onclick="submitForm()">Submit</button> will trigger the submitForm() function when clicked.

3. How is the combined information displayed to the user?

The combined information can be displayed using innerText or innerHTML to update a specific element on the page. For example, document.getElementById("output").innerText = fullInfo; shows the data dynamically.

4. How can you implement validation to ensure that the email field contains a valid email address?

You can use the type="email" attribute in the HTML <input> tag, which checks for proper email format. Additionally, in JavaScript, a regular expression (regex) can be used for more advanced email validation.

5. How can you restrict the input in the mobile number field to accept only numeric values?

In HTML, set type="number" or use inputmode="numeric" in the <input> tag. In JavaScript, you can also use isNaN() or a regex to ensure only digits are accepted in the field.

Lab Exercise-7. *(20/02/2025)*

Q :- Create XML file to store student info. like Enrolment No., Name Mobile No. , Email Id?

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3. **JavaScript (JS)** is a powerful, lightweight programming language used to make web pages interactive. It enables dynamic content, animations, form validations, and real-time updates without needing to reload the page as a key component of web development .
4. **XML (Extensible Markup Language)** is a flexible, text-based format used for storing and transporting data. It defines custom tags to structure data in a human-readable and machine-parsable way. XML is widely used in web services (SOAP), configuration files, and data exchange between systems. Unlike HTML, it focuses on data storage, presentation.

1. Index.html File

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```html
  <title>Student List</title>
  <link rel="stylesheet" href="style.css" />
 </head>
 <body>
  <h2>Student Information</h2>
  <table>
   <thead>
    <tr>
     <th>Enrolment Number</th>
     <th>Name</th>
     <th>Mobile Number</th>
     <th>Email ID</th>
    </tr>
   </thead>
   <tbody id="studentTable"></tbody>
  </table>
  <script src="script.js"></script>
 </body>
</html>
```

2. style.css File

```css
body {
  font-family: Arial, sans-serif;
  text-align: center;
  margin: 20px;
  padding-top: 5%;
  align-items: center;
}
```

```css
table {
   width: 80%;
   margin: auto;
   border-collapse: collapse;
   margin-top: 20px;
}
th, td {
   border: 2px solid black;
   padding: 10px;
   text-align: left;
}
```

3. student.xml File

```xml
<?xml version="1.0" encoding="UTF-8"?>
<Students>
   <Student>
      <EnrolmentNumber>12214802722</EnrolmentNumber>
      <Name>Yash Pandey</Name>
      <MobileNumber>9876543210</MobileNumber>
      <EmailId>yash.pandey@example.com</EmailId>
   </Student>

   <Student>
      <EnrolmentNumber>11114802722</EnrolmentNumber>
      <Name>Divyam Jain</Name>
      <MobileNumber>9123456789</MobileNumber>
      <EmailId>divyam.jain@example.com</EmailId>
   </Student>
```

```xml
  <Student>
    <EnrolmentNumber>10514802722</EnrolmentNumber>
    <Name>Shaswat Satyam</Name>
    <MobileNumber>9988776655</MobileNumber>
    <EmailId>shaswat.satyam@example.com</EmailId>
  </Student>

  <Student>
    <EnrolmentNumber>11614802722</EnrolmentNumber>
    <Name>Amit Singhal</Name>
    <MobileNumber>8765432109</MobileNumber>
    <EmailId>amit.singhal@example.com</EmailId>
  </Student>

  <Student>
    <EnrolmentNumber>12614802722</EnrolmentNumber>
    <Name>Sambhav Singhal</Name>
    <MobileNumber>9234567890</MobileNumber>
    <EmailId>sambhav.singhal@example.com</EmailId>
  </Student>

  <Student>
    <EnrolmentNumber>12914802722</EnrolmentNumber>
    <Name>Priyanshu Tiwari</Name>
    <MobileNumber>9012345678</MobileNumber>
    <EmailId>priyanshu.tiwari@example.com</EmailId>
  </Student>

</Students>
```

4. script.js File

```
document.addEventListener("DOMContentLoaded", async function () {
 await loadXML();
});
async function loadXML() {
 try {
  let response = await fetch("students.xml"); // Fetch XML file
  let xmlText = await response.text(); // Convert response to text
  let parser = new DOMParser();
  let xmlDoc = parser.parseFromString(xmlText, "text/xml"); // Parse XML
  displayData(xmlDoc);
 } catch (error) {
  console.error("Error loading XML:", error);
 }
}
function displayData(xml) {
 let table = document.getElementById("studentTable");
 let students = xml.getElementsByTagName("Student");

 for (let student of students) {
  let row = document.createElement("tr");

  let enrolmentCell = document.createElement("td");
  enrolmentCell.textContent =
   student.getElementsByTagName("EnrolmentNumber")[0].textContent;

  let nameCell = document.createElement("td");
  nameCell.textContent = student.getElementsByTagName("Name")[0].textContent;
```

```
    let mobileCell = document.createElement("td");
    mobileCell.textContent =
     student.getElementsByTagName("MobileNumber")[0].textContent;

    let emailCell = document.createElement("td");
    emailCell.textContent =
     student.getElementsByTagName("EmailId")[0].textContent;

    row.appendChild(enrolmentCell);
    row.appendChild(nameCell);
    row.appendChild(mobileCell);
    row.appendChild(emailCell);

    table.appendChild(row);
  }
}
```

OUTPUT :-

Student Information

Enrolment Number	Name	Mobile Number	Email ID
12214802722	Yash Pandey	9876543210	yash.pandey@example.com
11114802722	Divyam Jain	9123456789	divyam.jain@example.com
10514802722	Shaswat Satyam	9988776655	shaswat.satyam@example.com
11614802722	Amit Singhal	8765432109	amit.singhal@example.com
12614802722	Sambhav Singhal	9234567890	sambhav.singhal@example.com
12914802722	Priyanshu Tiwari	9012345678	priyanshu.tiwari@example.com

Viva Questions (Lab Exercise -7)

1. What is the root element of the XML document?

The root element is the top-level element that wraps all other elements in the XML file. For example, if the file contains student data, <students> would typically be the root element.

2. What elements are used to represent each student's information?

Each student's information is usually enclosed within a <student> tag, and inside it, tags like <name>, <roll>, <email>, or <grade> are used to hold specific data fields.

3. How can you add another student's information to this XML file?

To add a new student, simply append another <student> block within the root element <students>, including all required child elements like name, roll number, etc.

4. How can you modify an existing student's information in this XML file?

You can open the XML file in a text or XML editor and locate the specific <student> element. Then, directly edit the content inside the relevant child tags such as <name> or <email>.

5. How can you remove a student's information from this XML file?

To remove a student's data, locate their <student> element block and delete it entirely from the file. This includes all nested tags like name, roll, and other details.

Lab Exercise-8. *(20/03/2025)*

Q :- Write a php script to read data from txt file and display it in HTML table (the file contains data as Name: Password: Email)?

The **PHP (Hypertext Preprocessor)** is a widely-used open-source server-side scripting language designed for web development. It allows developers to create dynamic web pages, handle form data, interact with databases, and generate HTML content efficiently. PHP runs on various platforms and is compatible with multiple databases like **MySQL, PostgreSQL, and MongoDB**.

PHP, when combined with HTML and a database, enables the creation of dynamic and interactive web applications. PHP processes user requests on the server, generates HTML content, and interacts with databases like MySQL to store and retrieve data. Forms in HTML can send user inputs to a PHP script, which then processes and saves them in the database. PHP can also fetch stored data and dynamically update web pages, making it ideal for building login systems, e-commerce platforms, and content management systems. With seamless database connectivity using MySQLi or PDO, PHP ensures efficient data handling and secure transactions in web applications.

1. Data.txt File

data.txt ×

data.txt

```
Yash Pandey: 12214802722: yash@gmail.com
Divyam Jain: 11114802722: divyam@gmail.com
Shaswat Satyam: 10514802722: shaswat@gmail.com
Amit Singhal: 11614802722: amit@gmail.com
Priyanshu Saha: 09714802722: priyanshu@gmail.com
```

2. index.php File

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Users Data Table</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;      background-color: #f9f9f9;
    }
    table {
      width: 60%;      margin: 30px auto;
      border-collapse: collapse;      background-color: #fff;
      box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.1);
    }
  </style>
</head>
<body>
  <h2>Users Information Table</h2>
  <table>
    <tr>
      <th>Name</th>
      <th>Password</th>
      <th>Email</th>
    </tr>
```

```
        <?php
        $file = "data.txt";
        if (file_exists($file)) {
            $lines = file($file, FILE_IGNORE_NEW_LINES | FILE_SKIP_EMPTY_LINES);
            foreach ($lines as $line) {
                $parts = explode(":", $line);
                if (count($parts) == 3) {
                    echo "<tr><td>" . trim($parts[0]) . "</td><td>" . trim($parts[1]) . "</td><td>" .
                                            trim($parts[2]) . "</td></tr>";
                }
            }
        }       ?>
    </table>
</body>
</html>
```

OUTPUT :-

Users Information Table

Name	Password	Email
Yash Pandey	12214802722	yash@gmail.com
Divyam Jain	11114802722	divyam@gmail.com
Shaswat Satyam	10514802722	shaswat@gmail.com
Amit Singhal	11614802722	amit@gmail.com
Priyanshu Saha	09714802722	priyanshu@gmail.com

Viva Questions (Lab Exercise -8)

1. **How does the PHP script open the text file for reading?**

The PHP script uses the fopen() function with the mode "r" to open a file for reading. For example, fopen("data.txt", "r"); opens the file named data.txt in read-only mode.

2. **What function is used to split each line into an array using ":" as the delimiter?**

The explode(":", $line) function is used to split the string at each colon. It returns an array with values separated by the : symbol in the text line.

3. **What does the feof() function do?**

The feof() function checks if the end of the file has been reached. It's typically used in loops to continue reading the file until no more lines are left.

4. **How can you improve the script to handle empty lines or lines with incorrect formatting in the text file?**

You can use trim() to skip empty lines and count() to check if the split array has the expected number of elements before processing the data further.

5. **How can you add CSS styling to the HTML table to improve its appearance?**

You can use internal or external CSS with properties like border, padding, background-color, and text-align. For example, table { border-collapse: collapse; } td { padding: 8px; border: 1px solid #ccc; }.

Lab Exercise-9. *(20/03/2025)*

Q :- WAP for login authentication Using PHP. Design an html form which takes name and password from user and validate for stored username and password in file.?

The **PHP (Hypertext Preprocessor)** is a widely-used open-source server-side scripting language designed for web development. It allows developers to create dynamic web pages, handle form data, interact with databases, and generate HTML content efficiently. PHP runs on various platforms and is compatible with multiple databases like **MySQL, PostgreSQL, and MongoDB**.

PHP, when combined with HTML and a database, enables the creation of dynamic and interactive web applications. PHP processes user requests on the server, generates HTML content, and interacts with databases like MySQL to store and retrieve data. Forms in HTML can send user inputs to a PHP script, which then processes and saves them in the database. PHP can also fetch stored data and dynamically update web pages, making it ideal for building login systems, e-commerce platforms, and content management systems. With seamless database connectivity using MySQLi or PDO, PHP ensures efficient data handling and secure transactions in web applications.

1. LoginPage.html File

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="utf-8">
   <meta name="viewport" content="width=device-width, initial-scale=1">
   <title>Login Authentication With PHP</title>
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/css/"rel="stylesheet">
</head>
```

```
<body>
  <div class="container mt-5">
    <h2>Login Authentication With PHP</h2>
    <form action="login.php" method="POST">
      <div class="mb-3">
        <label for="inputName" class="form-label">UserName</label>
        <input type="text" class="form-control" id=" inputName " required>
      </div>
      <div class="mb-3">
        <label for="inputPassword" class="form-label">Password</label>
        <input type="password" class="form-control" id="inputPassword" required>
      </div>
      <button type="submit" class="btn btn-primary">Login</button>
    </form>
  </div>
</body>
</html>
```

2. Users.txt File

users.txt ×

users.txt

```
1  yashpandey:12214802722
2  divyamjain:11114802722
3  shaswatsatyam:10514802722
4  amitsinghal:11614802722
5  priyanshusaha:09714802722
```

3. login.php File

```
<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
  $username = trim($_POST["username"]);
  $password = trim($_POST["password"]);
  $file = "users.txt";
  if (file_exists($file)) {
    $users = file($file, FILE_IGNORE_NEW_LINES | FILE_SKIP_EMPTY_LINES);
    foreach ($users as $user) {
      list($storedUser, $storedPass) = explode(":", $user);
      if ($username === trim($storedUser) && $password === trim($storedPass)) {
        $_SESSION["username"] = $username;
        header("Location: welcome.php");
        exit;
      }
    }
  }
  }
?>
```

4. welcome.php File

```
<?php
session_start();
if (!isset($_SESSION["username"])) {
  header("Location: LoginPage.html");
  exit;
}
?>
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Welcome Page After Login Authentication</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; margin-top: 100px; }
  </style>
</head>
<body>
  <h2>Welcome, <?php echo htmlspecialchars($_SESSION["username"]); ?>!</h2>
  <a href="logout.php">Logout</a>
</body>
</html>
```

5. logout.php File

```
<?php
        session_start();
        session_destroy();
        header("Location: LoginPage.html");
        exit;
?>
```

OUTPUT (Before Login):-

Login Authentication With PHP!

UserName

yashpandey

Password

12214802722

We'll never share your Email & Password with anyone else.

Submit

OUTPUT (Post Login):-

Welcome, yashpandey!

Logout

Viva Questions (Lab Exercise -9)

1. How does the PHP script retrieve the username and password entered by the user from the HTML form?

The PHP script retrieves the username and password using the $_POST superglobal. For example, $_POST['username'] gets the username, and $_POST['password'] gets the password entered in the form.

2. How does the script read the stored usernames and passwords from the file?

The script uses the fopen() function to open the file, then reads it line by line using fgets(). Each line typically contains a username and password, which are processed for authentication.

3. What function is used to split each line of the file into an array using ":" as the delimiter?

The explode(":", $line) function splits each line into an array by the colon : delimiter. This allows the script to separate the username and password stored in each line.

4. How does the script display a message indicating whether the login attempt was successful or not?

The script compares the entered username and password with the values read from the file. If they match, it outputs a success message like echo "Login successful!"; otherwise, it displays echo "Invalid login credentials.".

5. How can you enhance the security of the login process?

You can enhance security by hashing the passwords using functions like password_hash() and password_verify() in PHP. Additionally, using HTTPS, salting passwords, and limiting login attempts can further secure the login process.

Lab Exercise-10. *(20/03/2025)*

Q :- Write PHP Script for storing & retrieving user info. from MySQL table?

1. **Design A HTML page which takes Name, Address, Email and Mobile No. From user (Sends Request To register.php)**
2. **Store this data in Mysql database / text file.**
3. **Next page display all user in html table using PHP (display.php)**

The **PHP (Hypertext Preprocessor)** is a widely-used open-source server-side scripting language designed for web development. It allows developers to create dynamic web pages, handle form data, interact with databases, and generate HTML content efficiently. PHP runs on various platforms and is compatible with multiple databases like **MySQL, PostgreSQL, and MongoDB**.

MySQL is a widely used **relational database management system (RDBMS)** that seamlessly integrates with PHP to build dynamic and data-driven web applications. PHP provides built-in support for MySQL through extensions like **MySQLi (MySQL Improved)** and **PDO (PHP Data Objects)**, allowing developers to connect to databases, execute queries, and manage data efficiently.

1. LoginPage.html File

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="utf-8">
   <meta name="viewport" content="width=device-width, initial-scale=1">
   <title>Login Authentication With PHP And MySQL</title>
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css" rel="stylesheet">
</head>
```

```html
<body>
  <div class="container mt-5">
    <h2>  Login Authentication With PHP!</h2>
    <form action="register.php" method="POST">
      <div class="row mb-3">
        <div class="col-md-4">
          <label class="form-label">UserName</label>
          <input type="text" class="form-control" id="inputName">
        </div>
        <div class="col-md-4">
          <label class="form-label">Email</label>
          <input type="email" class="form-control" id="inputEmail">
        </div>
      </div>
      <div class="row mb-3">
        <div class="col-md-4">
          <label class="form-label">Address</label>
          <input type="text" class="form-control" id="inputAddress">
        </div>
        <div class="col-md-4">
          <label class="form-label">Phone</label>
          <input type="number" class="form-control" id="inputNumber">
        </div>
      </div>
      <button type="submit" class="btn btn-primary">Submit</button>
    </form>
  </div>
</body>
</html>
```

2. setup.sql File

```
CREATE DATABASE user_db;
USE user_db;

CREATE TABLE users (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(255),    address VARCHAR(255),
   email VARCHAR(255),    mobile VARCHAR(15)
);
```

3. register.php File

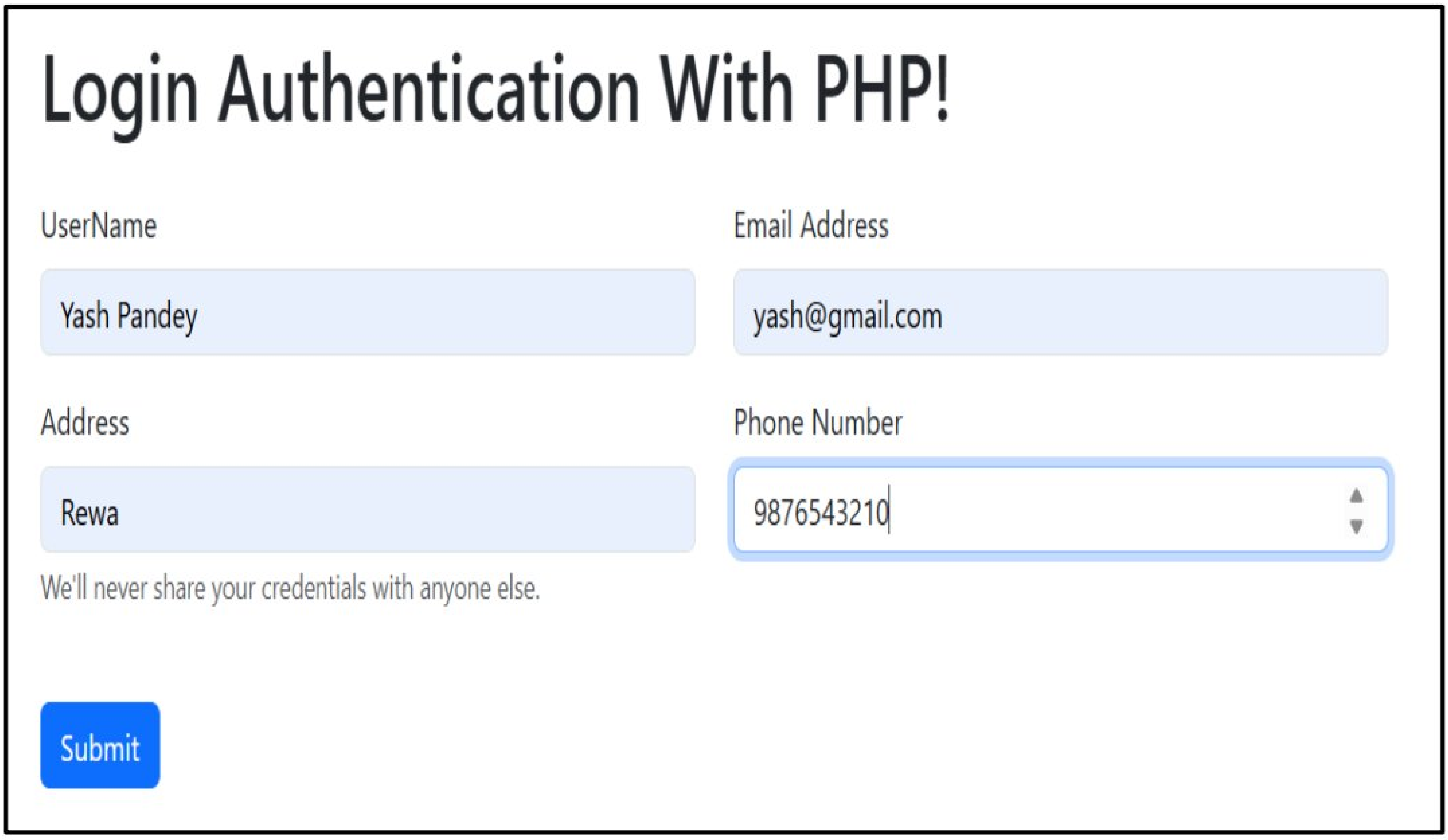
```
<?php
$servername="localhost";
$username="root";
$password="ROOT4321";
$dbname="user_db";
$conn=new mysqli($servername,$username,$password,$dbname);
if($conn->connect_error){die("Connection failed: ".$conn->connect_error);}
$name=$_POST['name'];        $address=$_POST['address'];
$email=$_POST['email'];        $mobile=$_POST['mobile'];
$sql="INSERT INTO users(name,address,email,mobile)
VALUES('$name','$address','$email','$mobile')";

if($conn->query($sql)===TRUE){
   echo"User Registered Successfully!<br>";
   echo"<a href='display.php'>View Users</a>";
}else{echo"Error: ".$conn->error;}
$conn->close();
?>
```

4. display.php File

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "user_db";
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
$sql = "SELECT * FROM users";
$result = $conn->query($sql);
?>

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Users Lists</title>
</head>
<body>
  <div class="container mt-5">
    <h2>Registered Users</h2>
    <table class="table table-bordered">
      <tr>
        <th>Name</th>          <th>Address</th>
        <th>Email</th>          <th>Mobile No.</th>
      </tr>
```

```
<?php if ($result->num_rows > 0): ?>
  <?php while ($row = $result->fetch_assoc()): ?>
    <tr>
      <td><?= $row['name'] ?></td>          <td><?= $row['address'] ?></td>
      <td><?= $row['email'] ?></td>          <td><?= $row['mobile'] ?></td>
    </tr>
  <?php endwhile; ?>
<?php else: ?>
  <tr><td colspan="4">No users found</td></tr>
<?php endif; ?>
</table>
</div>
</body>
</html>
```

OUTPUT (Before Form Submision):-



OUTPUT (After Form Submision):-

User Registered Successfully!

View Users

(Clicking "View Users" redirects to display.php)

OUTPUT (On display.php File):-

Registered Users

Name	Address	Email	Mobile No.
Yash Pandey	Rewa	yash@gmail.com	9876543210
Divyam Jain	New Delhi	divyam@gmail.com	8765432109
Amit Singhal	Roorkee	amit@gmail.com	7654321098
Shaswat Satyam	Patna	shaswat@gmail.com	857432185

Viva Questions (Lab Exercise -10)

1. **Explain the process of connecting PHP to a MySQL database.**

To connect PHP to a MySQL database, you use the mysqli_connect() function, providing the server address, username, password, and database name. If the connection is successful, it returns a connection object; otherwise, it shows an error message.

2. **How do you create a MySQL database and table for storing user information?**

To create a MySQL database, use the CREATE DATABASE SQL command. For creating a table, use CREATE TABLE, specifying the column names and data types (e.g., username, password, email). You can execute these commands through PHP using mysqli_query().

3. **How do you insert user information into a MySQL table using PHP?**

To insert user information, use the INSERT INTO SQL query. You prepare the query in PHP with user input data, then execute it using the mysqli_query() function to insert the data into the MySQL table.

4. **Explain the use of the mysqli query() function in PHP for executing MySQL queries.**

The mysqli_query() function is used to send SQL queries to the MySQL database. It can be used for SELECT, INSERT, UPDATE, and DELETE operations. It returns TRUE for successful queries and FALSE for errors.

5. **How can you secure user input before storing it in a MySQL database using PHP?**

To secure user input, use prepared statements with mysqli_prepare() and parameterized queries to prevent SQL injection. Additionally, use mysqli_real_escape_string() to escape special characters and ensure safe input.

Lab Exercise-11. *(10/04/2025)*

(Beyond Syllabus , Additional Question – 1)

Q :- WAP To Design An Simple Calculator Using HTML , CSS And JavaScript ?

A webpage is a digital document displayed on the internet through a web browser, consisting of elements like text, images, videos, and links. It is created using HTML for structure and CSS for styling, making it either static with fixed content or dynamic with interactive features.

1. **HTML (HyperText Markup Language**) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and functional webpages for users.
2. **CSS (Cascading Style Sheets)** is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly .
3. **JavaScript (JS)** is a powerful, lightweight programming language used to make web pages interactive. It enables dynamic content, animations, form validations, and real-time updates without needing to reload the page as a key component of web development .

1. Index.html File

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Calculator-App</title>
```

```
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet" />

  <link rel="stylesheet" href="style.css" />

</head>

<body>

  <div class="container mt-5">

    <h1 class="text-center mb-4">Easy-to-Use calculator</h1>

    <div class="calculator bg-light p-3">

      <input type="text" class="form-control mb-2" id="Text-Field" disabled />

      <div class="row g-2">

        <!-- Top row: AC, C, x², / -->

        <div class="col-3"><button class="btn btn-light"
onclick="ClearText()">AC</button></div>

        <div class="col-3"><button class="btn btn-light"
onclick="ClearLastCharacter()">C</button></div>

        <div class="col-3"><button class="btn btn-light"
onclick="SquareValue()">x<sup>2</sup></button></div>

        <div class="col-3"><button class="btn btn-light"
onclick="AddNumber('/')">/</button></div>


        <!-- Digits and operations -->

        ${[7,8,9,'*',4,5,6,'-',1,2,3,'+'].map(v=>`<div class="col-3"><button class="btn btn-light"
onclick="AddNumber('${v}')">${v}</button></div>`).join('')}


        <!-- Bottom row: 0, 00, ., = -->

        <div class="col-3"><button class="btn btn-light"
onclick="AddNumber(0)">0</button></div>

        <div class="col-3"><button class="btn btn-light"
onclick="AddNumber('00')">00</button></div>

        <div class="col-3"><button class="btn btn-light"
onclick="AddNumber('.')">.</button></div>

        <div class="col-3"><button class="btn btn-primary"
onclick="Evaluate()">=</button></div>
```

```html
    </div>
   </div>
   <h2 class="text-center mt-3 mx-2">  Made with ♡ by Yash Pandey!</h2>
  </div>
  <script src="script.js"></script>
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></scrip
t>
</body>
</html>
```

2. Style.css File

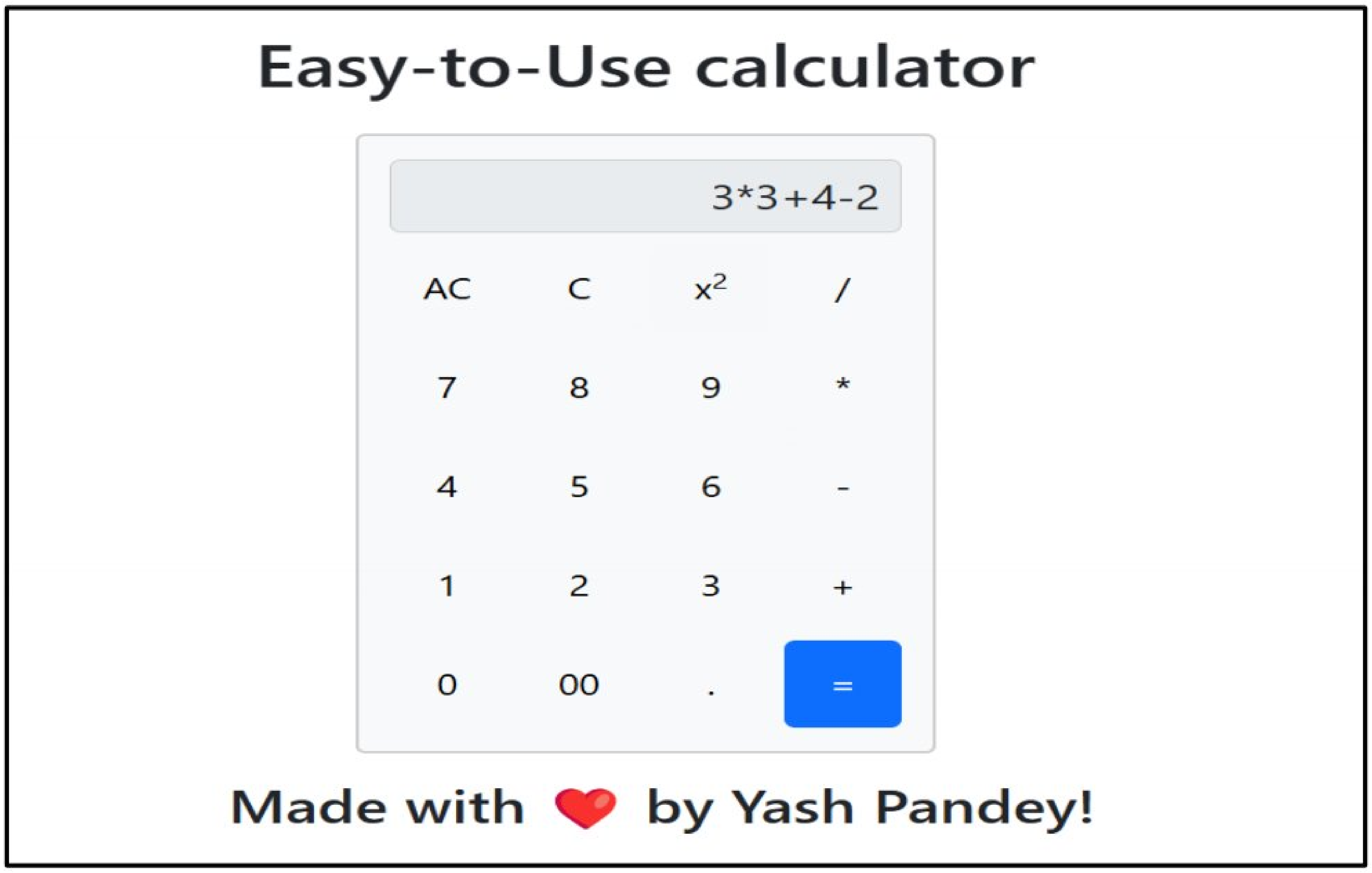
```css
.calculator {
  max-width: 300px;
  margin: auto;
  border: 2px solid #ccc;
  padding: 10px;
  border-radius: 5px;
}
.btn-custom {
  width: 100%;
  height: 60px;
  font-size: 20px;
}
.display {
  width: 100%;            height: 50px;
  text-align: right;      font-size: 24px;
  margin-bottom: 10px;
  border: 1px solid #ccc;
}
```

3. Script.js File

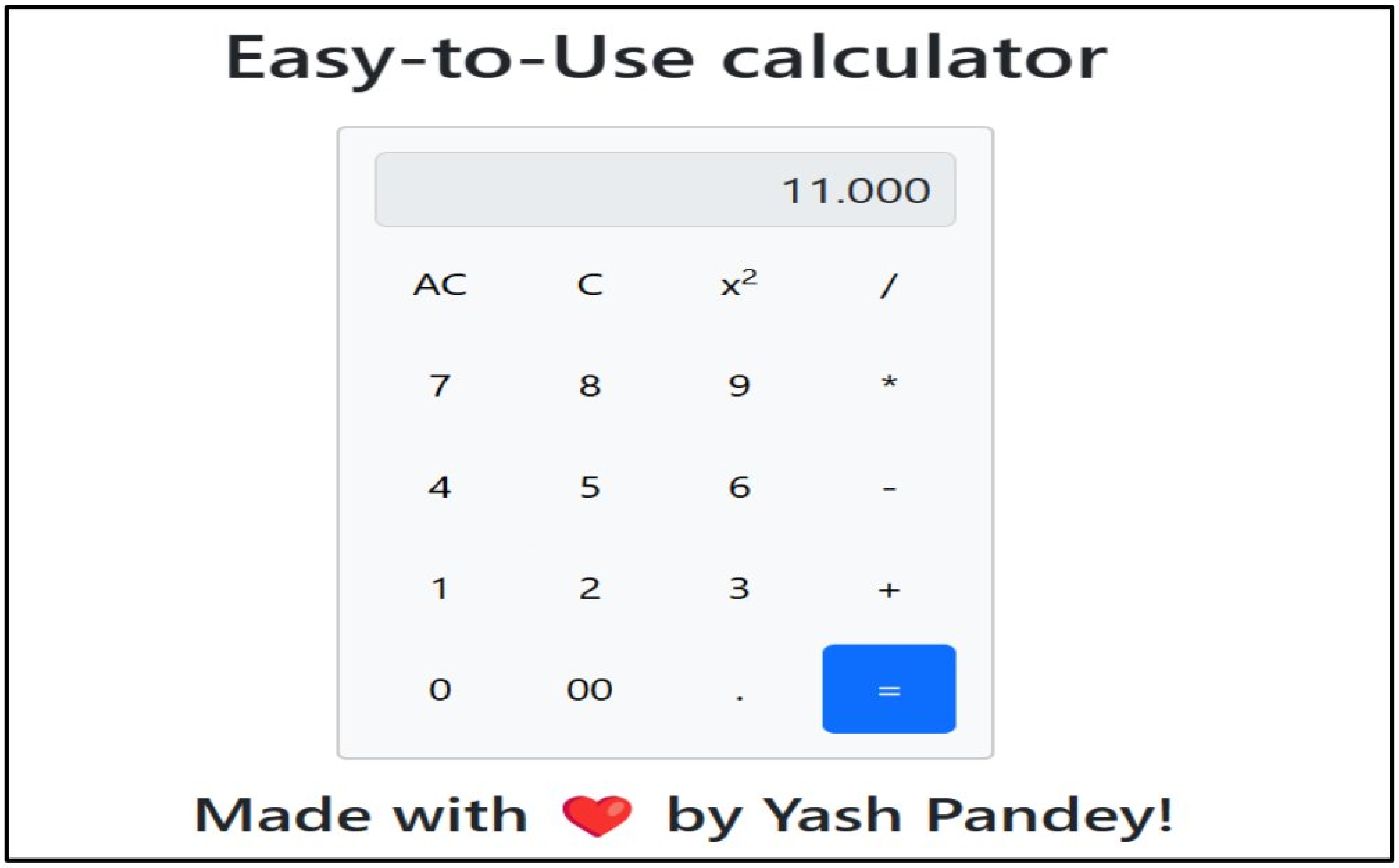
```
const inputFormTextArea = document.getElementById("Text-Field");


function AddNumber(num) {
  inputFormTextArea.value += num;
}


function ClearText() {
  inputFormTextArea.value = "";
}


function ClearLastCharacter() {
  inputFormTextArea.value = inputFormTextArea.value.slice(0, -1);
}


function Evaluate() {
  inputFormTextArea.value = evaluateExpression(inputFormTextArea.value);
}
function SquareValue() {
  const val = evaluateExpression(inputFormTextArea.value);
  inputFormTextArea.value = evaluateExpression(val * val);
}
function evaluateExpression(expr) {
  try {
    return eval(expr).toFixed(3);
  } catch {
    return "Error in expression";
  }
}
```

Live Working Link -> https://yashpandey1405.github.io/simple-calculator-app/

Output (Before Evaluation) :



Output (After Evaluation) :



Viva Questions (Beyond Syllabus -1)

1. **How would you structure the HTML for the calculator?**

To create the HTML structure for the calculator, you would use a combination of <div> elements for layout and <button> elements for digits and operators. The display area can be created using an <input> element of type="text" to show the result.

2. **What is the role of CSS in styling the calculator?**

CSS is used to style the calculator's layout, buttons, and display area. It helps align the buttons in a grid format, apply colors to buttons, and style the display area to make it visually appealing and functional. For example, you can use display: grid for button layout and background-color for button styling.

3. **How does JavaScript handle user input for the calculator?**

JavaScript handles user input by attaching onclick events to each button. When a button is clicked, it triggers a function that appends the value to the display or performs an arithmetic operation based on the operator clicked. The value of the display is updated dynamically after each input.

4. **How do you perform arithmetic operations like addition and subtraction in JavaScript?**

Arithmetic operations in JavaScript are performed using standard operators like + for addition and - for subtraction. The input values are retrieved from the display area, converted into numbers, and then the operation is applied. The result is then displayed back on the webpage.

5. **What would you do to prevent division by zero in the calculator?**

To prevent division by zero, you can check if the denominator is zero before performing the division. In the JavaScript function handling the division, add a condition like if (denominator === 0) { alert('Cannot divide by zero'); } to display an error message instead of performing the operation.

Lab Exercise-12. *(10/04/2025)*

(Beyond Syllabus , Additional Question – 2)

Q :- WAP To Design An Simple Quiz Webpage Using HTML , CSS And JavaScript ?

A webpage is a digital document displayed on the internet through a web browser, consisting of elements like text, images, videos, and links. It is created using HTML for structure and CSS for styling, making it either static with fixed content or dynamic with interactive features.

1. **HTML (HyperText Markup Language**) forms the backbone of a webpage, using tags like <h1>, <p>, and <a> to define headings, paragraphs, and links. Together, HTML and CSS create visually appealing and functional webpages for users.
2. **CSS (Cascading Style Sheets)** is a styling language used to enhance the appearance of webpages by controlling their layout and design. It allows developers to define styles for elements like colors, fonts, spacing, and positioning, making webpages visually appealing and user-friendly .
3. **JavaScript (JS)** is a powerful, lightweight programming language used to make web pages interactive. It enables dynamic content, animations, form validations, and real-time updates without needing to reload the page as a key component of web development .

1. Index.html File

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
```

```html
    rel="stylesheet"
    integrity="sha384-QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhjY6hW+ALEwIH"
    crossorigin="anonymous"
  />
  <link
    rel="stylesheet"
    href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.2/css/all.min.css"
  />
  <link rel="stylesheet" href="style.css" />
  <title>Web Tech Quiz</title>
 </head>
 <body class="bg-light">
  <nav
    class="navbar navbar-expand-lg bg-dark border-bottom border-body p-3"
    data-bs-theme="dark"
  >
    <div
      class="container-fluid d-flex justify-content-between align-items-center"
    >
      <a class="navbar-brand text-white" href="#"
        ><i class="fas fa-feather-alt fa-xl" style="color: #1da1f2"></i>
          ChirpNest</a
      >
      <h5 class="text-white mx-auto">
        Welcome To ChirpNest! Let's test your web skills!
      </h5>
    </div>
  </nav>
  <div class="container mt-5">
```

```html
<h1>Web Tech Quiz!</h1>
<form id="form-input">
 <div class="row">
  <div class="col mb-3">
   <label for="firstName" class="form-label">First Name</label
   ><input type="text" class="form-control" id="firstName" />
  </div>
  <div class="col mb-3">
   <label for="lastName" class="form-label">Last Name</label
   ><input type="text" class="form-control" id="lastName" />
  </div>
 </div>
 <div class="row">
  <div class="col mb-3">
   <label for="Email" class="form-label">Email</label
   ><input type="text" class="form-control" id="Email" />
   <div id="emailHelp" class="form-text">
    We'll never share your email.
   </div>
  </div>
  <div class="col mb-3">
   <label for="Number" class="form-label">Phone</label
   ><input type="number" class="form-control" id="Number" />
  </div>
 </div>
 <button type="submit" class="btn btn-primary">Submit</button>
</form>
<hr />
<h2 class="text-center mt-5 mb-3" id="quiz-header"></h2>
```

```html
<div id="quiz" style="display: none; margin-top: 30px">
 <div class="row mt-5">
  <div class="col-md-6">
   <p>1. HTML hyperlink tag?</p>
   <input type="radio" name="q1" value="1" /> &lt;a&gt;
   <input type="radio" name="q1" value="2" /> &lt;link&gt;
   <input type="radio" name="q1" value="3" /> &lt;href&gt;
   <input type="radio" name="q1" value="4" /> &lt;hyperlink&gt;
  </div>
  <div class="col-md-6">
   <p>2. CSS background color?</p>
   <input type="radio" name="q2" value="1" /> background-color
   <input type="radio" name="q2" value="2" /> color
   <input type="radio" name="q2" value="3" /> bgcolor
   <input type="radio" name="q2" value="4" /> background
  </div>
 </div>
 <div class="row mt-5">
  <div class="col-md-6">
   <p>3. HTML 'id' attribute?</p>
   <input type="radio" name="q3" value="1" /> Defines style
   <input type="radio" name="q3" value="2" /> Unique element
   <input type="radio" name="q3" value="3" /> Links CSS class
   <input type="radio" name="q3" value="4" /> New element
  </div>
  <div class="col-md-6">
   <p>4. JS comment?</p>
   <input type="radio" name="q4" value="1" /> /* comment */
   <input type="radio" name="q4" value="2" /> // comment
```

```html
    <input type="radio" name="q4" value="3" /> &lt;!-- comment --&gt;
    <input type="radio" name="q4" value="4" /> # comment
  </div>
</div>
<div class="row mt-5">
  <div class="col-md-6">
    <p>5. HTML paragraph?</p>
    <input type="radio" name="q5" value="1" /> &lt;p&gt;
    <input type="radio" name="q5" value="2" /> &lt;h1&gt;
    <input type="radio" name="q5" value="3" /> &lt;div&gt;
    <input type="radio" name="q5" value="4" /> &lt;span&gt;
  </div>
  <div class="col-md-6">
    <p>6. &lt;div&gt; display?</p>
    <input type="radio" name="q6" value="1" /> inline
    <input type="radio" name="q6" value="2" /> block
    <input type="radio" name="q6" value="3" /> flex
    <input type="radio" name="q6" value="4" /> grid
  </div>
</div>
<div class="row mt-5">
  <div class="col-md-6">
    <p>7. External CSS link?</p>
    <input type="radio" name="q7" value="1" /> &lt;link&gt;
    <input type="radio" name="q7" value="2" /> &lt;external&gt;
    <input type="radio" name="q7" value="3" /> &lt;css&gt;
    <input type="radio" name="q7" value="4" /> &lt;script&gt;
  </div>
  <div class="col-md-6">
```

```
    <p>8. HTML image tag?</p>
    <input type="radio" name="q8" value="1" /> &lt;image&gt;
    <input type="radio" name="q8" value="2" /> &lt;img&gt;
    <input type="radio" name="q8" value="3" /> &lt;picture&gt;
    <input type="radio" name="q8" value="4" /> &lt;src&gt;
  </div>
</div>
<div class="row mt-5">
  <div class="col-md-6">
    <p>9. HTML 'class'?</p>
    <input type="radio" name="q9" value="1" /> Links CSS
    <input type="radio" name="q9" value="2" /> Specific element
    <input type="radio" name="q9" value="3" /> User class
    <input type="radio" name="q9" value="4" /> Element type
  </div>
  <div class="col-md-6">
    <p>10. CSS text color?</p>
    <input type="radio" name="q10" value="1" /> color
    <input type="radio" name="q10" value="2" /> font-color
    <input type="radio" name="q10" value="3" /> text-color
    <input type="radio" name="q10" value="4" /> background-color
  </div>
</div>
<div class="row mt-5">
  <div class="col-md-6">
    <p>11. JS class select?</p>
    <input type="radio" name="q11" value="1" />
    document.getElementById('example')
    <input type="radio" name="q11" value="2" />
```

```
        document.querySelector('.example')
        <input type="radio" name="q11" value="3" />
        document.querySelector('#example')
        <input type="radio" name="q11" value="4" />
        document.getElementByClass('example')
      </div>
      <div class="col-md-6">
        <p>12. JS button click?</p>
        <input type="radio" name="q12" value="1" /> onchange
        <input type="radio" name="q12" value="2" /> onclick
        <input type="radio" name="q12" value="3" /> onhover
        <input type="radio" name="q12" value="4" /> onfocus
      </div>
    </div>
    <button class="btn btn-success mt-4 mb-2" id="submitQuiz">
      Submit Quiz
    </button>
  </div>
  <h2 class="mb-3 text-success" id="quiz-score"></h2>
  <h4 class="mt-2 mb-3 text-danger" id="quiz-incorrect"></h4>
</div>
<footer class="footer text-center bg-dark text-white py-3">
  <div class="container">
    <div
      class="mb-3 d-flex justify-content-center align-items-center gap-3"
    >
      <a
        href="https://github.com/YashPandey1405/"
        class="text-white"
```

```
          target="_blank"
          ><i class="fab fa-github fa-xl"></i></a
        ><a
          href="https://www.linkedin.com/in/yashpandey29/"
          class="text-white"
          target="_blank"
          ><i class="fab fa-linkedin fa-xl"></i
        ></a>
      </div>
      <p class="mb-1">&copy; YashPandey Private Limited</p>
    </div>
  </footer>
  <script src="script.js"></script>
  <script
    src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
    integrity="sha384-
YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDzOxhy9GkcIdslK1eN7N6jIeHz"
    crossorigin="anonymous"
  ></script>
 </body>
</html>
```

2. Style.css File

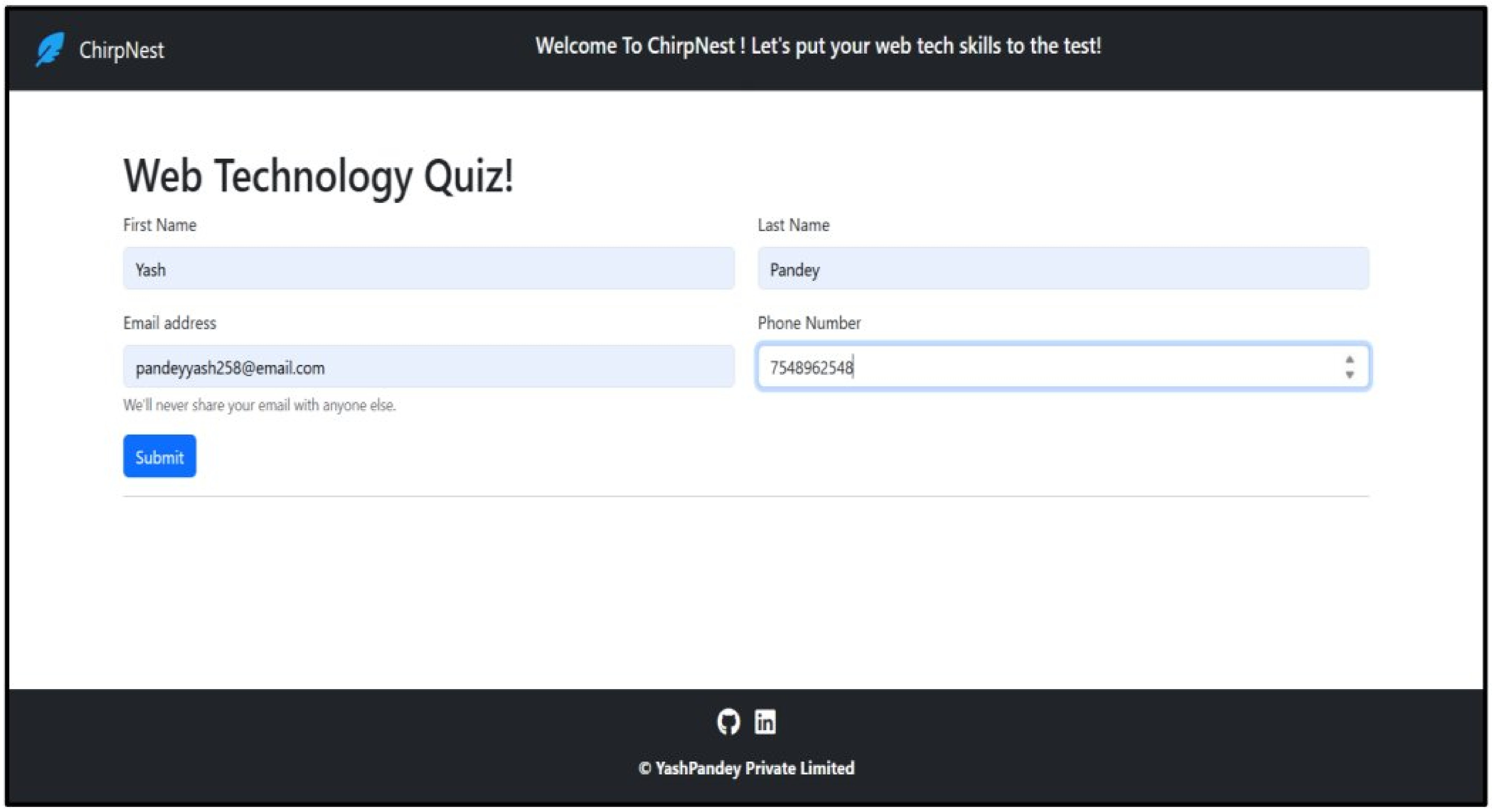
```
html,
body {
  height: 100%;
  display: flex;
  flex-direction: column;
}
```

3. Script.js File

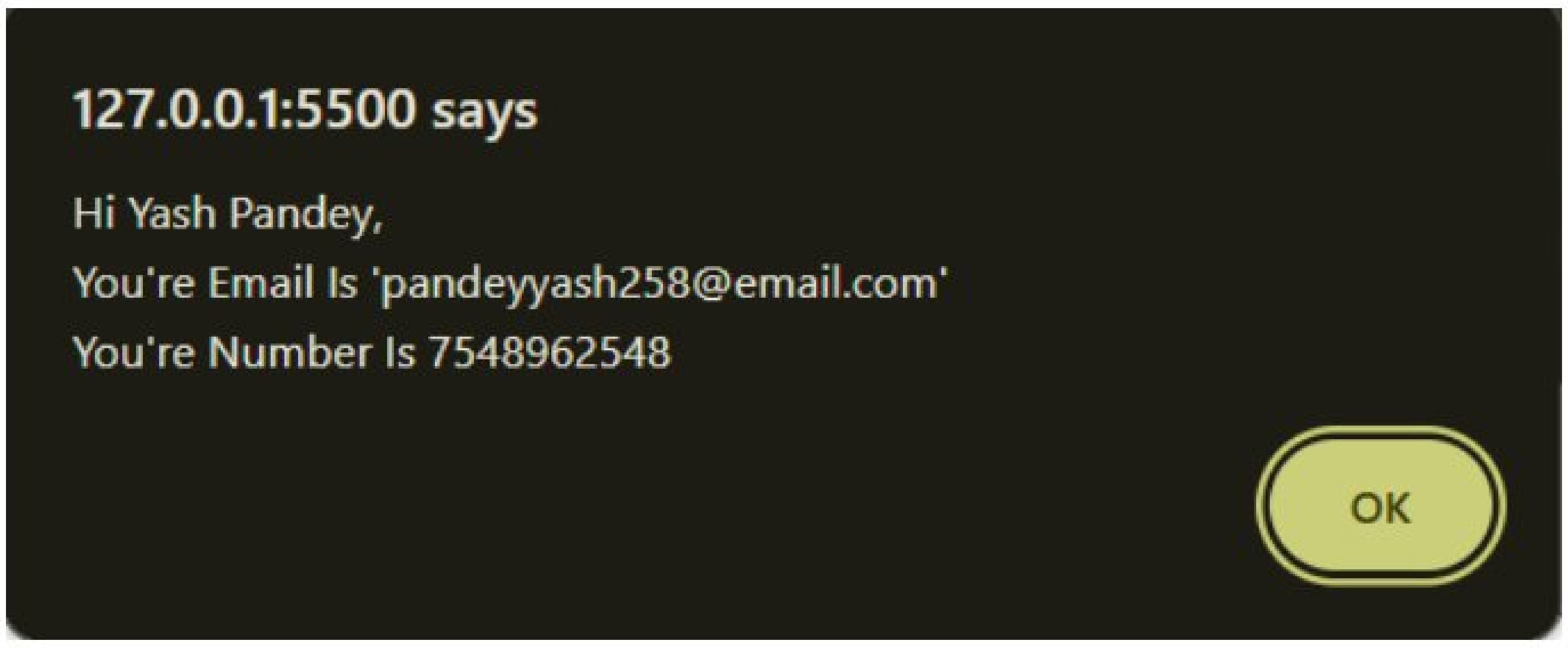
```
const inputFormDetais = document.getElementById("form-input");

const quizHeader = document.getElementById("quiz-header");

const quizScore = document.getElementById("quiz-score");

const quizIncorrect = document.getElementById("quiz-incorrect");


let userName = "John", incorrectAnswers = "";


document.getElementById("submitQuiz").addEventListener("click", e => {
  e.preventDefault();   let score = 0;
  for (let i = 1; i <= 12; i++) {
    const userAnswer = document.querySelector(`input[name="q${i}"]:checked`);
    if (userAnswer && userAnswer.value === answers[`q${i}`]) score++;
    else incorrectAnswers += `Question ${i}: You answered ${userAnswer?.value}, correct:
${answers[`q${i}`]}.\n`;
  }
  alert(`Hi ${userName}, score: ${score}/12`);
  quizScore.innerText = "Incorrect Solutions:";
  quizIncorrect.innerText = incorrectAnswers;
});
inputFormDetais.addEventListener("submit", e => {
  const firstName = document.getElementById("firstName").value, lastName =
document.getElementById("lastName").value,
     email = document.getElementById("Email").value, number =
document.getElementById("Number").value;
  userName = `${firstName} ${lastName}`;
  alert(`Hi ${userName}, Email: '${email}', Number: ${number}`);
  document.getElementById("quiz").style.display = "block";
  quizHeader.innerText = `Hi ${userName}, Web Tech Quiz`;
});
```

Live Working Link -> https://yashpandey1405.github.io/web-tech-quiz/

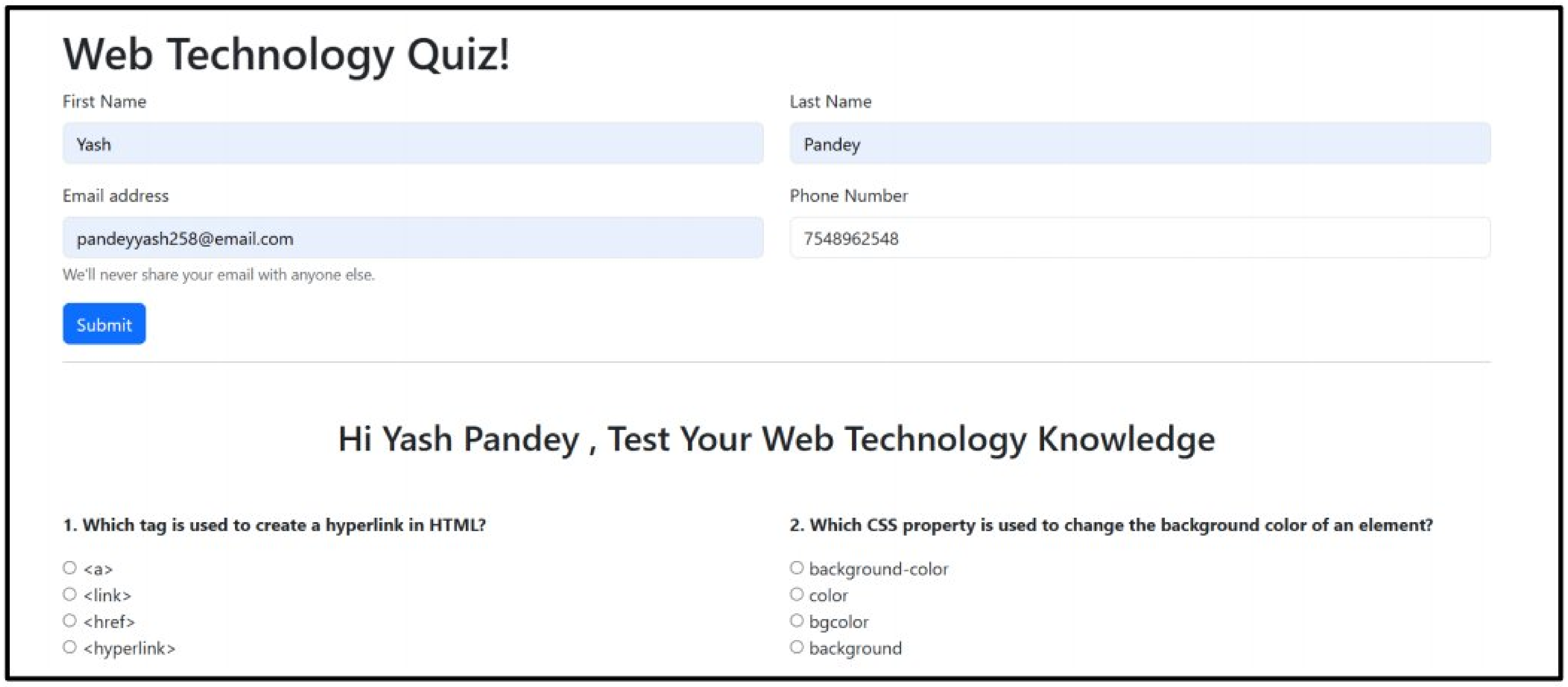
Output (Fill Details Before Actual Quiz) :

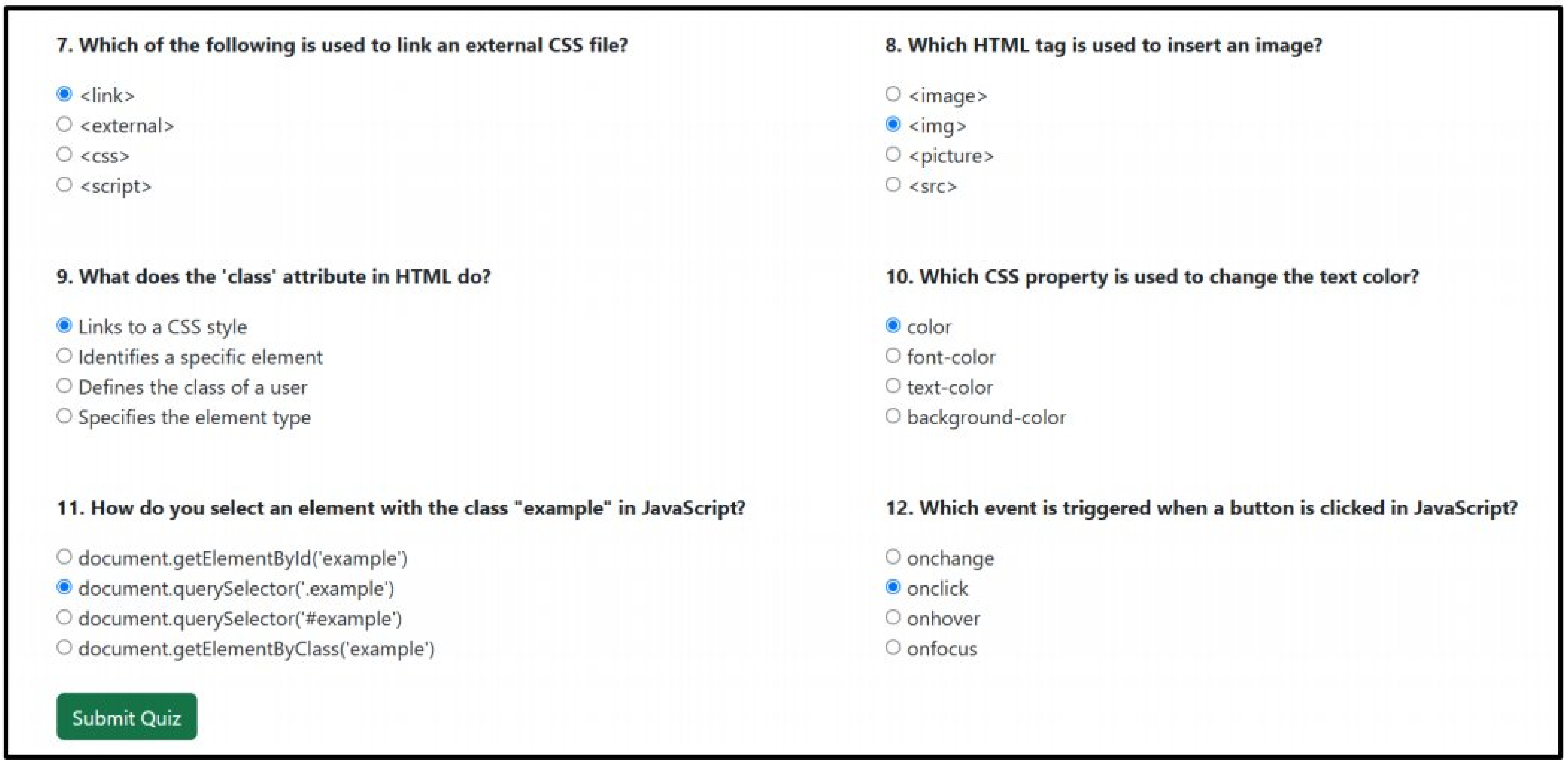


Output (Welcome Message POST Form Submit) :



Output (Before Submit Of Quiz) :





Output (After Submit Of Quiz) :

Viva Questions (Beyond Syllabus -2)

1. How would you structure the HTML for the quiz?

The HTML for the quiz would consist of a set of questions, each with multiple-choice options, which can be created using <div>, <p>, and <input> elements. Each question would have radio buttons for answer options, allowing the user to select one answer per question. The "Submit" button To make score calculation.

2. What is the role of CSS in styling the quiz page?

CSS is used to style the quiz page to make it visually appealing and easy to navigate. It helps in organizing the layout of questions and answers, styling the radio buttons, setting background colors, and making the "Submit" button stand out. You can use flexbox or grid to align the questions and answers properly.

3. How does JavaScript handle the quiz answers and calculate the score?

JavaScript collects the selected answers using querySelectorAll for radio buttons. It compares each selected answer with the correct answer, counts the correct ones, and displays the score after the "Submit" button is clicked.

4. How do you prevent the user from submitting the quiz without answering all the questions?

To prevent submission without answering all questions, you can add validation in JavaScript. Before calculating the score, check if all radio buttons have been selected. If any question is left unanswered, show an alert or display a message asking the user to answer all the questions before submitting.

5. How would you display the quiz score to the user after submission?

The score is calculated by comparing the user's selected answers with the correct ones. Once the comparison is done, the score is dynamically displayed on the webpage by updating the content of an element (like a div or paragraph) to show the total score after the user clicks the "Submit" button.